

**[REAL-TIME MALAYSIAN TRAFFIC SIGN DETECTION AND ALERT
SYSTEM]**

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**[REAL-TIME MALAYSIAN TRAFFIC SIGN DETECTION AND ALERT
SYSTEM]**

[KISHENTHIRAN A/L SAGADEVAN]

This report is submitted in partial fulfillment of the requirements for the
Bachelor of [Computer Science (Artificial Intelligence)] with Honours.

FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY
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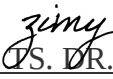
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I hereby declare that this project report entitled
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DEDICATION

I dedicate this project to my beloved parents, whose unconditional love, endless sacrifices, and constant prayers have been the foundation of my strength throughout this academic journey. Their unwavering belief in me has been my greatest motivation during times of self-doubt and fatigue. I am also deeply grateful to my siblings and extended family members who have stood by me with encouragement, patience, and warmth, never failing to lift my spirit when I needed it most.

This work is also dedicated to my close friends, who have supported me with their kindness, advice, and positivity — making even the most challenging moments bearable. To every student or individual who works tirelessly in silence, balancing responsibilities, expectations, and ambition — this is for you. May this project serve as a symbol of resilience, growth, and determination.

Lastly, I dedicate this achievement to my future self — as a reminder that with focus, effort, and the right intentions, anything is possible.

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Lastly, I am truly grateful to my family for their constant encouragement, patience, and emotional support throughout this academic journey. Their belief in me has been the foundation of my perseverance and success.

ABSTRACT

Traffic signs are vital to road safety, yet drivers can miss or misread them under real-world conditions. This project presents a real-time Malaysian traffic-sign detection and alert system and introduces two new contributions beyond a standard driver-assist prototype: (i) an authority-facing JKR Dashboard and (ii) an explainable visibility score that quantifies image quality issues. YOLOv8 is fine-tuned for sign detection and a lightweight CNN (ResNet-18) classifies speed-limit signs; detection results are rendered with bounding boxes and confidence, and critical signs trigger visual/audio alerts. Each event is logged with timestamp and annotated frame. For JKR, the dashboard provides filters (date/label/problem), a “Top-20 worst by visibility” list, one-click CSV/XLSX export, and suggested actions. The visibility score combines blur (variance of Laplacian), contrast (grayscale standard deviation), and glare (saturated-pixel ratio) into a 0–100 metric with explicit thresholds, allowing repeatable, auditable triage of poor-visibility locations. A curated dataset assembled from Kaggle, Roboflow, and academic sources was used; low-support or low-performing classes were pruned after evaluation to improve robustness. Results show stable convergence of the detector and 94% accuracy for the speed-limit classifier. Significantly, for drivers proactive alerts reduce missed signs; for JKR, structured logs and an interpretable quality metric turn detections into actionable evidence for enforcement and maintenance planning on Malaysian roads.

ABSTRAK

Tanda jalan ialah elemen penting keselamatan jalan raya, namun pemandu boleh terlepas pandang atau tersalah tafsir dalam keadaan pemanduan sebenar. Projek ini membangunkan sistem pengesanan dan amaran tanda jalan Malaysia secara masa nyata serta memperkenalkan dua sumbangan baharu berbanding prototaip bantuan pemandu biasa: (i) Papan Pemuka JKR untuk pihak berkuasa dan (ii) skor keterlihatan berpenjelasan yang mengukur isu kualiti imej. YOLOv8 ditala halus untuk pengesanan manakala CNN ringan (ResNet-18) mengelaskan tanda had laju; keputusan pengesanan dipaparkan bersama kotak sempadan dan keyakinan, dan tanda kritikal mencetuskan amaran visual/audio. Setiap kejadian dilogkan dengan cap masa serta imej beranotasi. Untuk JKR, papan pemuka menyediakan penapis (tarikh/label/masalah), senarai “20 terburuk mengikut keterlihatan”, eksport CSV/XLSX satu klik dan cadangan tindakan. Skor keterlihatan menggabungkan kabur (varian Laplacian), kontras (sisihan piawai kelabu) dan silau (nisbah piksel tepu) kepada metrik 0–100 dengan ambang yang jelas, membolehkan triage yang boleh diaudit bagi lokasi ber-keterlihatan rendah. Set data berkurasi daripada Kaggle, Roboflow dan sumber akademik digunakan; kelas berprestasi rendah atau kurang data digugurkan selepas penilaian untuk menambah keteguhan. Keputusan menunjukkan penumpuan penalaan yang stabil dan ketepatan 94% bagi pengelas had laju. Ia menjadi signifikan untuk pemandu, amaran proaktif mengurangkan tanda yang terlepas; untuk JKR, log berstruktur dan metrik kualiti yang boleh diterangkan menukar pengesanan kepada bukti yang boleh ditindak bagi penguatkuasaan dan perancangan penyenggaraan di jalan raya Malaysia.

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LIST OF ABBREVIATIONS

FYP	-	Final Year Project
AI	-	Artificial Intelligence
ADAS	-	Advanced Driver Assistance Systems
GPS	-	Global Positioning System
ITS	-	Intelligent Transportation System
YOLO	-	You Only Look Once
CCA	-	Canonical Correlation Analysis
CNN	-	Convolutional Neural Network
ResNet-18	-	Residual Network with 18 layers
VSSA-NET	-	Visual Saliency and Spatial Attention Network
SVM	-	Support Vector Machine
HOG	-	Histogram of Oriented Gradients
HSV	-	Hue, Saturation, Value (color space)
PCA	-	Principal Component Analysis
LDA	-	Linear Discriminant Analysis
KNN	-	K-Nearest Neighbors
CRISP-DM	-	Cross-Industry Standard Process for Data Mining
Faster R-CNN	-	Faster Region-based Convolutional Neural Network
DFD	-	Data Flow Diagram

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Chapter 1: INTRODUCTION

1.1 Introduction

Traffic signs are crucial components of road safety infrastructure, guiding drivers with speed limits, warnings, and navigational cues. However, many accidents in Malaysia are attributed to drivers failing to notice or misinterpreting traffic signs, often due to distractions, fatigue, poor visibility, or unfamiliar roads. While Advanced Driver Assistance Systems (ADAS) are gaining popularity, most existing systems are trained on foreign datasets and do not adequately support local Malaysian road environments.

The lack of a high-quality, annotated Malaysian traffic sign dataset poses a significant challenge in training accurate and reliable deep learning models for this purpose. Moreover, real-time traffic sign detection demands high processing efficiency, robust classification, and seamless communication between detection modules and the user interface — all under varying lighting and environmental conditions. Traditional solutions like dashcams or navigation apps provide passive support and cannot offer the proactive assistance required to reduce violations and accidents.

To address these gaps, this project investigates, designs, and develops a real-time Malaysian traffic sign detection and alert system using deep learning. The system not only provides real-time visual and auditory alerts to drivers but also logs detected signs for post-drive review by authorities such as JKR (Jabatan Kerja Raya). The system architecture is also designed to support future integration into ADAS, contributing to intelligent transportation systems and smarter roads in Malaysia.



Figure 1.1 Example of Malaysian Traffic Signs

1.2 Problem Statements

Traffic signs are essential for maintaining road safety by providing critical instructions and warnings to drivers. However, in real-world driving conditions, many signs are missed, ignored, or misunderstood due to driver distraction, fatigue, adverse weather, poor visibility, or unfamiliar routes. This issue is especially problematic in Malaysia, where road signs can vary in design and placement. These overlooked signs often lead to traffic violations or accidents, highlighting the need for a more proactive system that can assist drivers in real time.

While dashcams and navigation apps like Waze are widely used, they primarily offer passive support or depend on crowdsourced or GPS data, which may not detect temporary or newly installed traffic signs. Dashcams merely record footage and require manual review, offering no preventive feedback to the driver. Existing ADAS systems like Mobileye are powerful but are often not optimized for Malaysian traffic signs, as they are trained on foreign datasets and require expensive hardware setups.

There is a gap for a localized, real-time traffic sign detection system that not only alerts drivers but also records sign encounters for further analysis.

Furthermore, traffic enforcement and road safety monitoring rely heavily on physical presence (e.g., speed traps) or manual analysis of dashcam footage after incidents occur. As of right now, there isn't an automated system that can record interactions with traffic signs in real time to help JKR confirm compliance, identify accident-prone locations, or spot dangerous driving practices. Drivers, JKR, and ADAS developers would all benefit from a smart detection and alert system that is camera-based, lightweight, and tailored for Malaysian road conditions. This would improve proactive safety and accountability after an incident.

1.3 Objective

- i. **To investigate** YOLOv8 performance for detecting Malaysian traffic signs from real-time video under varied lighting and weather.
- ii. **To design** an AI pipeline—YOLOv8 detector + ResNet-18 (CNN) classifier—that delivers real-time driver alerts and timestamped logs for JKR analysis.
- iii. **To develop** a modular real-time prototype with Driver and JKR dashboards, ready for integration into ITS/ADAS.

1.4 Scope

The system will consist of the following four functional modules:

1. Traffic Sign Detection Module
 - Train and fine-tune a deep learning model specifically on Malaysian traffic signs to accurately detect signs from real-time video feeds.
2. Traffic Sign Classification Module
 - Automatically classify each detected sign into its respective classes (e.g., Speed limit 50, Bicycle Lane, Children, No Parking).
3. Alert and Notification Module
 - Provide immediate visual and audio alerts to the driver when a traffic sign is detected, improving on-road awareness and safety.
4. Dashboard Interface and Logging Module
 - Displays detections in real time and stores logs (with timestamps) for post-analysis. These logs can later support JKR in enforcement, road safety audits, and infrastructure planning.

1.5 Project Contribution

1.5.1 Target Users (Stakeholders)

- i. **Drivers** - To improve safety by alerting them to traffic signs they may miss due to fatigue, distraction, or poor visibility due to adverse weathers.
- ii. **JKR (Jabatan Kerja Raya)** – Gain access to structured detection logs for monitoring road safety compliance, accident hotspot analysis, and planning sign maintenance.
- iii. **ADAS & Smart Vehicle Developers** - To integrate Malaysian sign detection into intelligent vehicles.

1.5.2 Expected Outcomes

The expected outcome of this project is a functional prototype of a real-time traffic sign detection and alert system tailored to Malaysian road conditions. The system will process live video input from a dash-mounted camera, detect and classify local traffic signs, and issue real-time alerts to the driver. It will also include a logging feature that records detection events and timestamps, offering valuable data for driver compliance analysis and review by JKR (Jabatan Kerja Raya).

In addition to enhancing road safety for drivers, the system aims to support JKR in monitoring violations, analyzing accident-prone zones, and planning road safety improvements, while also serving as a plug-and-play module for future ADAS integrations. Ultimately, the system will contribute to safer, smarter roads and help modernize Malaysia's approach to intelligent transportation management.

1.6 Organization of the Thesis

This thesis is structured into six chapters: introduction, literature review and methodology, requirement analysis, system design, implementation and results, and finally, conclusion and future work.

The **first chapter** introduces the background of the project, outlines the problem statements, objectives, target users, scope, and expected outcomes of the project titled *Real-Time Malaysian Traffic Sign Detection and Alert System*. This chapter also summarizes the overall research structure and goals.

The **second chapter** covers both the literature review and the project methodology. The literature review explores existing works in the domain of traffic sign detection, computer vision techniques, object detection models like YOLO, and related systems implemented in both commercial and academic contexts. The methodology section explains the project development framework, which is guided by the CRISP-DM (Cross-Industry Standard Process for Data Mining) model. Each phase of CRISP-DM — from business understanding, data understanding, data preparation, modelling, evaluation, and deployment — is described in the context of this project.

The **third chapter** presents the system's requirement analysis. It details both functional and non-functional requirements, identifies user needs, and describes the environment and constraints considered during system planning.

The **fourth chapter** focuses on the system design. It discusses the high-level architecture of the system, internal module design (including detection, classification, alerting, and logging), and the layout of the dashboard interface used to display real-time sign information and notifications.

The **fifth chapter** illustrates the implementation process and results. It presents the actual development of the prototype, showcases detection outputs, visual alert samples, and includes quantitative evaluation of the model's accuracy and performance. Discussions analyze the system's effectiveness across various real-world scenarios.

The **final chapter** concludes the project by summarizing key outcomes, reflecting on limitations, and proposing future improvements. Suggestions are provided for scaling the system for wider ADAS integration or smart enforcement applications.

1.7 Summary

This project proposes a real-time Malaysian Traffic Sign Detection and Alert System designed to enhance road safety through deep learning-based visual recognition. The system aims to assist drivers in avoiding accidents caused by missing or misinterpreted traffic signs due to distraction, fatigue, or poor visibility. By leveraging camera input and object detection models, the system provides live visual and audio alerts to drivers while also logging sign detections for review by traffic enforcement.

The project objectives include investigating the effectiveness of deep learning models for Malaysian traffic sign recognition, designing a real-time detection and alert interface, and developing a prototype that supports both driver awareness and compliance analysis. The system architecture is also designed with future adaptability in mind, allowing integration into ADAS (Advanced Driver Assistance Systems).

Target users include private and commercial drivers, traffic police, and ADAS developers. The project is expected to produce a modular, localized detection system that improves driver decision-making, supports traffic law enforcement, and contributes to the development of intelligent transportation systems in Malaysia.

CHAPTER 2: LITERATURE REVIEW AND PROJECT METHODOLOGY

2.1 Introduction

Traffic sign detection is a critical field in intelligent transportation systems and Advanced Driver Assistance Systems (ADAS). Accurate and timely recognition of traffic signs help in enhancing driver awareness, reducing accidents, and enabling autonomous or semi-autonomous driving functions. As road infrastructure becomes more complex and vehicle automation advances, the need for robust traffic sign detection systems that perform under diverse conditions grows stronger. Numerous research studies and commercial systems have explored various approaches, including deep learning models, handcrafted features, color segmentation, and hybrid methods.

This chapter presents a comprehensive review of related work in two main parts: existing academic research (Section 2.2.1) and commercial or practical implementations (Section 2.2.2). The review covers object detection architectures such as YOLO, transformer-enhanced models, classical SVM-HOG methods, and real-time recognition systems. Strengths and limitations are discussed to identify current gaps in the field. Additionally, a comparison between existing solutions and the proposed system is outlined in Section 2.3 to justify the project's direction.

2.2 Facts/Findings

2.2.1 Domain

The domain of this project lies within the field of Intelligent Transportation Systems (ITS), a multidisciplinary area that combines traffic engineering, artificial intelligence, and data analytics to enhance transportation efficiency and safety. ITS focuses on creating smart solutions such as driver assistance systems, adaptive traffic control, and vehicle-to-infrastructure communication to reduce congestion, minimize accidents, and improve mobility. According to a bibliographic analysis by Linjing Li

et al. (2010), ITS research has been rapidly evolving, contributing significantly to advancements in real-time decision-making technologies, including traffic monitoring and automated recognition systems.

In the Malaysian context, traffic signs play a critical role in road safety, but are often overlooked due to driver distraction, poor weather conditions, or unfamiliar environments. Hence, developing an AI-based real-time traffic sign detection system can greatly assist drivers in staying alert and compliant with road regulations. Unlike standardized datasets such as GTSRB, Malaysia's traffic signs include multilingual, region-specific, and uniquely designed signage, which necessitates a localized detection model for accuracy and usability.

This project leverages YOLOv8 for fast object detection and ResNet18 for fine-grained classification of speed limit signs. These deep learning models enable the system to process live or recorded video footage in real-time, identify traffic signs, and issue visual or audio alerts. Such systems are not only valuable for individual drivers but also benefit road safety authorities, smart city initiatives, and autonomous vehicle development. The integration of AI into traffic management exemplifies how ITS contributes to reducing road fatalities and enhancing the overall driving experience (Li et al., 2010).

2.2.2 Existing Academic Research

This subsection reviews ten academic research papers related to traffic sign detection and recognition. These studies employ a wide range of techniques including deep learning (YOLO, CNNs, Transformers), traditional machine learning (SVM, HOG), and hybrid methods. Table 2.1 summarizes the key techniques used in each study, followed by detailed discussions of their contributions, advantages, and limitations.

Table 2-1 Summary of Techniques Used in Existing Academic Research

No	Title/Author	Techniques/Model Used	Purpose
1.	YOLO-CCA (Jiang et al., 2024)	YOLOv7 + Transformer + LCFE + GCFC	Context-aware traffic sign detection
2.	VSSA-NET (Yuan et al., 2019)	Vertical Spatial Sequence Attention Network	Small-size sign detection using spatial attention
3.	DFR-TSD (Ahmed et al., 2020)	Enhance-Net + CNN-based classification	Weather-resilient detection
4.	Total Recall (Saha et al., 2018)	Hierarchical Dilated CNN with Residual Blocks	High-accuracy, low-memory recognition
5.	SVM + HOG (Ahmed et al., 2021)	Histogram of Oriented Gradient + SVM	Sign detection in video for Bangladeshi dataset
6.	HOG + SVM (Tang et al., 2020)	Hand-crafted features + SVM classifier	Lightweight traffic sign classification
7.	Improved YOLOv3 (Liu & Xiong, 2019)	Dilated Convolution + Spatial Pyramid	Enhanced multi-scale feature detection
8.	Color Feature + Improved SVM (Zhou & Xiao, 2018)	Color segmentation + SVM	Detection using sign color and shape cues

9.	Statistical Learning (<i>Lillo-Castellano et al., 2015</i>)	HSV segmentation + PCA + LDA + KNN/SVM	Lightweight segmentation and classification
10.	Mexican Signs + Deep Learning (<i>Rodríguez et al., 2022</i>)	Faster R-CNN + CNN Classifier	Dataset-specific sign recognition with augmentation

After reviewing the ten selected papers, several key insights emerge regarding the advancement of traffic sign detection and classification. The research covers a broad spectrum of techniques, from traditional machine learning models like SVM with HOG features to more sophisticated deep learning frameworks such as YOLOv7 with Transformer-based augmentations. Recent models like YOLO-CCA and VSSA-NET emphasize spatial and contextual awareness to detect small and partially occluded signs, which is crucial for real-world deployments. Others, like DFR-TSD and Improved YOLOv3, focus on robustness against environmental challenges such as bad weather or complex backgrounds. Meanwhile, lightweight models such as HOG+SVM and color feature-based SVM aim to reduce computational overhead, enabling use in resource-constrained systems.

Informed by these findings, this project adopts a hybrid model—YOLOv8 for fast and accurate sign detection and a fine-tuned ResNet18 for detailed speed limit classification. This combination draws from the strengths of the reviewed models: the accuracy and real-time capabilities of YOLO-based detectors, and the specialized classification refinement seen in multi-stage approaches. Furthermore, unlike most prior research focused on generic or international datasets, this system is tailored for Malaysian road conditions, addressing a key gap identified across the literature.

2.2.2.1 YOLO-CCA: A Context-Based Approach for Traffic Sign Detection

This paper introduces a context-based algorithm utilizing YOLOv7 as the baseline model. It incorporates an adaptive local context feature enhancement (LCFE) module with multi-scale dilation convolution and a global context feature collection (GCFC) module. A Transformer-based context collection augmentation (CCA) module is also employed for superior multi-level feature fusion. (Jiang et al., 2024)

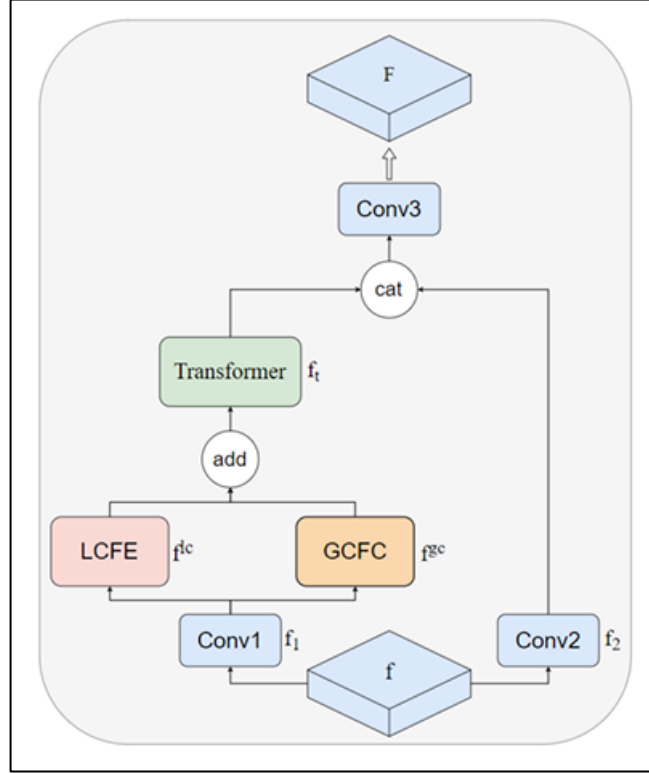


Figure 2.1 YOLO-CCA Architecture

The system achieves a mean Average Precision (mAP) of 92.1% on the Tsinghua-Tencent 100K dataset. It also improves detection accuracy with fewer parameters compared to standard YOLOv7 models, showing the potential of combining context-aware modules with efficient object detection frameworks.

However, the model's architecture is quite complex and may require substantial computational resources. Its high reliance on deep transformer components and multi-scale processing can hinder deployment on lightweight or embedded systems.

2.2.2.2 VSSA-NET: Vertical Spatial Sequence Attention Network for Traffic Sign Detection

The VSSA-NET architecture proposes a multi-resolution feature fusion network with densely connected deconvolution layers and skip connections. It reformulates traffic sign detection as a spatial sequence classification and regression problem. The model introduces a vertical spatial sequence attention (VSSA) module to enhance contextual understanding. (Yuan et al., 2019)

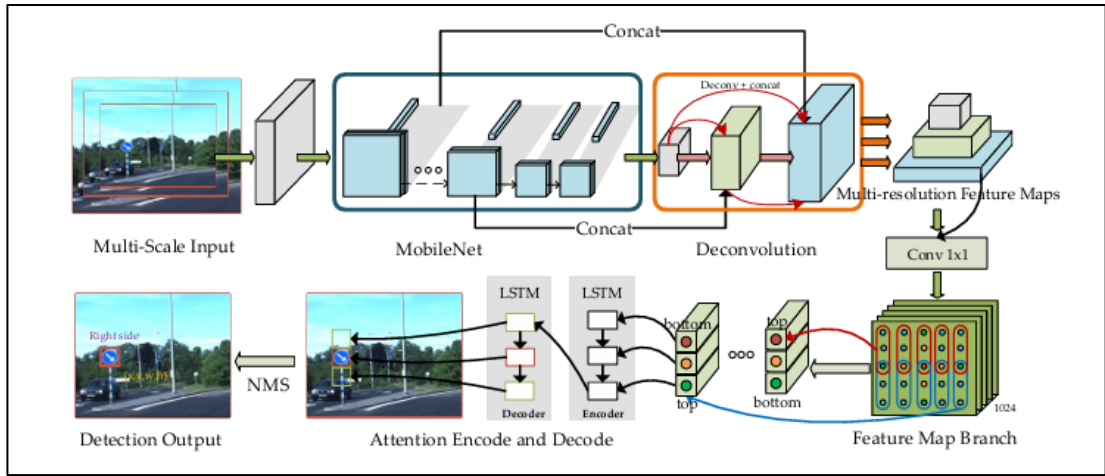


Figure 2.2 Architecture of VSSA-NET

This architecture effectively addresses the detection of small-sized traffic signs by incorporating contextual and spatial sequence information. It significantly improves detection performance, particularly in cluttered scenes where small signs are easy to overlook.

Nonetheless, the overall model is computationally demanding and may not be suitable for real-time applications without high-end processing units. The vertical attention and multiple deconvolution layers add significant processing overhead.

2.2.2.3 DFR-TSD: A Deep Learning Based Framework for Robust Traffic Sign Detection Under Challenging Weather Conditions

This framework includes a CNN-based challenge classifier, an encoder-decoder Enhance-Net for image enhancement, and separate CNN architectures for sign

detection and classification. It focuses on enhancing traffic sign regions in challenging images to improve detection accuracy. (Ahmed *et al.*, 2020)

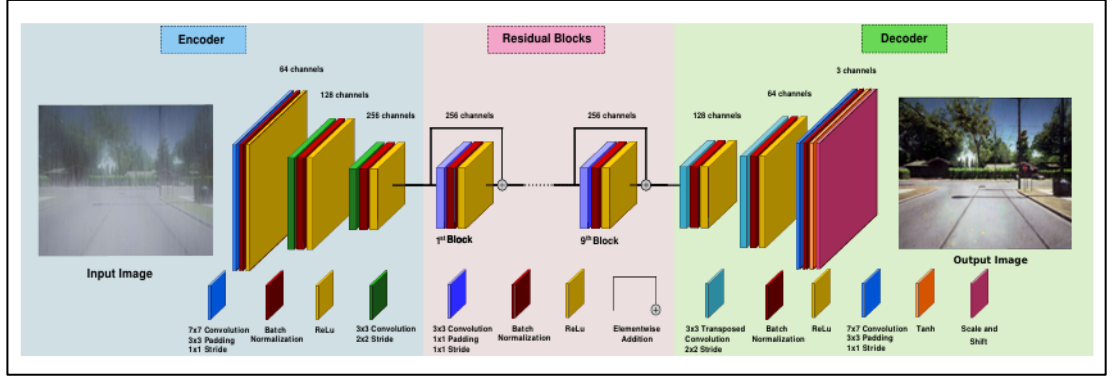


Figure 2.3 Architecture of DFR-TSD

It demonstrates improved precision and recall under adverse weather conditions such as rain, fog, or low-light environments. It also outperforms state-of-the-art object detection networks like Faster-RCNN and R-FCN in these scenarios.

However, the approach may require extensive training data representing different environmental conditions. Furthermore, the multi-network structure increases computational complexity, making it less efficient for real-time applications.

2.2.2.4 Total Recall: Understanding Traffic Signs using Deep Hierarchical Convolutional Neural Networks

This paper presents a novel architecture comprising residual convolutional blocks with hierarchical dilated skip connections. The model is designed to recognize traffic signs across multiple benchmarks with improved accuracy and low resource requirements. (Saha *et al.*, 2018)

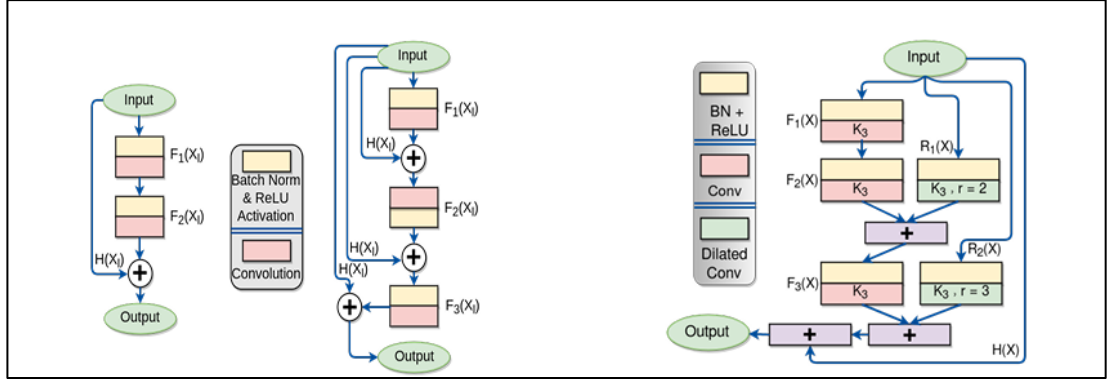


Figure 2.4 Architecture of Total Recall

The system achieves 99.33% accuracy on the German Traffic Sign Recognition Benchmark and demonstrates low memory and computational complexity. This makes it suitable for deployment on resource-constrained environments such as embedded systems.

However, the model's performance on non-German datasets, such as Malaysian road signs, remains untested. Additionally, while efficient, the architecture's real-time applicability may still be limited by inference delays depending on the hardware used.

2.2.2.5 Traffic Sign Detection and Recognition Model Using Support Vector Machine and Histogram of Oriented Gradient

This study proposes an automated approach for detecting and recognizing Bangladeshi traffic signs from video frames using Support Vector Machine (SVM) and Histogram of Oriented Gradient (HOG) features. (Ahmed et al., 2021)

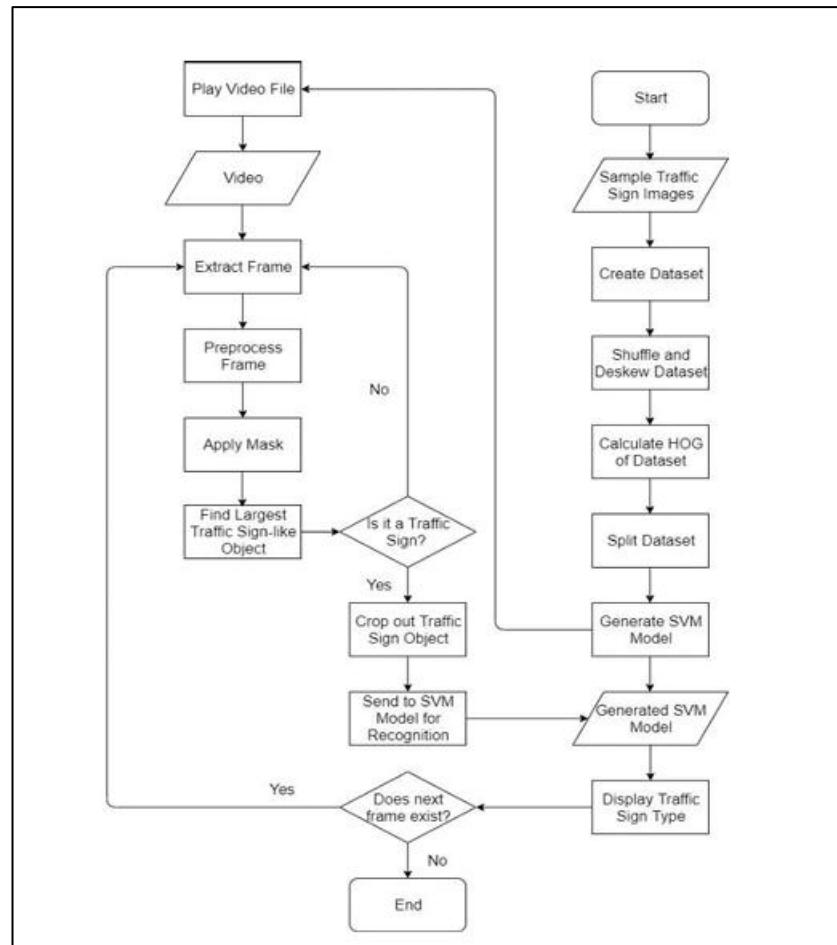


Figure 2.5 Flowchart of the SVM + HOG Techniques

The method achieves 96.15% accuracy with high precision and recall and effectively recognizes traffic signs from video input. It provides a strong baseline using traditional machine learning techniques in environments where deep learning may not be feasible.

However, the solution is tailored to Bangladeshi traffic signs and may not generalize well to other regional signs. It may also struggle with performance in poor lighting conditions or when signs are partially obscured.

2.2.2.6 Traffic Sign Recognition Based on HOG Feature and SVM

This research uses HOG for feature extraction and SVM for classification. The authors build a dataset with various categories of traffic signs and perform preprocessing to support the classification task. SVM (Tang et al., 2020)

The method boasts low computational complexity, making it ideal for real-time or embedded systems. It also delivers good accuracy on clean and high-quality image data using standard traffic sign shapes.

However, relying on hand-crafted features limits flexibility and adaptability compared to deep learning. Furthermore, this system focuses more on classification than real-time detection, and it cannot perform full pipeline detection in real-world scenarios.

2.2.2.7 A Real-Time Traffic Sign Detection Model Based on Improved YOLOv3

This paper improves YOLOv3 by using dilated convolutions and a three-layer spatial pyramid cascade structure for enhanced multi-scale feature fusion. (Liu & Xiong, 2019)

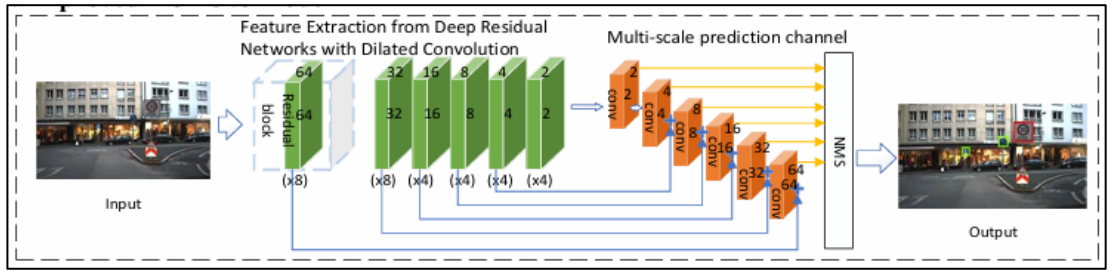


Figure 2.6 Architecture of YOLOv3

The model shows improved detection accuracy, especially for small and distant traffic signs. Despite the added features, it maintains fast inference speeds, making it suitable for real-time systems such as in-vehicle ADAS.

However, the improvements introduce higher model complexity than vanilla YOLOv3. This could result in increased memory and computation demand, especially on lower-end devices.

2.2.2.8 Traffic Sign Detection and Recognition Based on Color Features and Improved Support Vector Machine Algorithm

This paper proposes a detection and recognition system that utilizes the standardized color features of traffic signs to identify potential regions and an improved Support Vector Machine (SVM) algorithm for classification. The method is

based on detecting red or blue colors that commonly represent regulatory signs. (Zhou & Xiao, 2018)

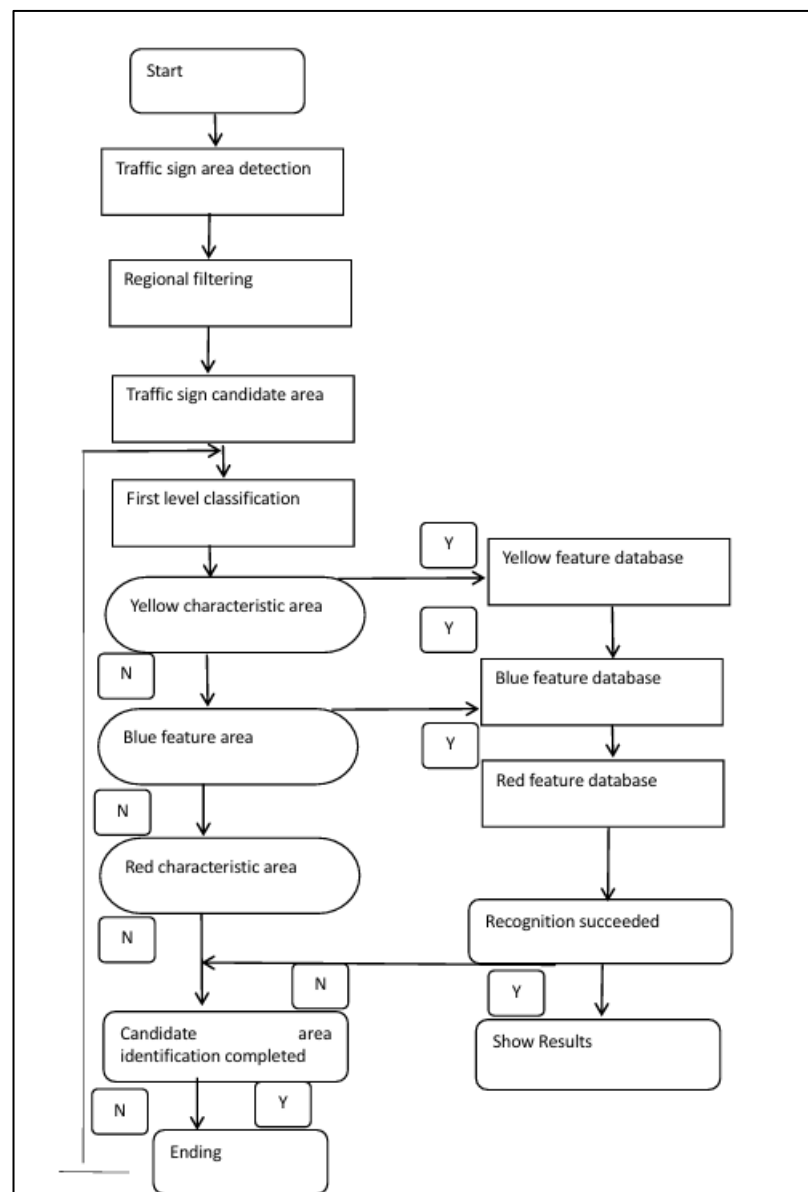


Figure 2.7 Flowchart of the Color Features + SVM Techniques

The system achieves a recognition performance of 97.03% across multiple traffic sign categories by combining effective color-based cues with optimized SVM classification techniques. It is also data-efficient compared to deep learning, making it viable in scenarios with limited training samples.

However, the system is highly dependent on color visibility. In conditions such as night driving, shadows, or faded signage, the performance significantly drops. It

also has limitations in detecting signs that are dirty, partially obscured, or vary in design from the training data.

2.2.2.9 Traffic Sign Segmentation and Classification Using Statistical Learning Methods

This study presents a two-phase system that begins with color segmentation in the HSV space and then filters signs based on their shape. In the classification phase, Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) are used for dimensionality reduction, followed by classifiers such as K-Nearest Neighbor (KNN) or Support Vector Machine (SVM). (Lillo-Castellano *et al.*, 2015)

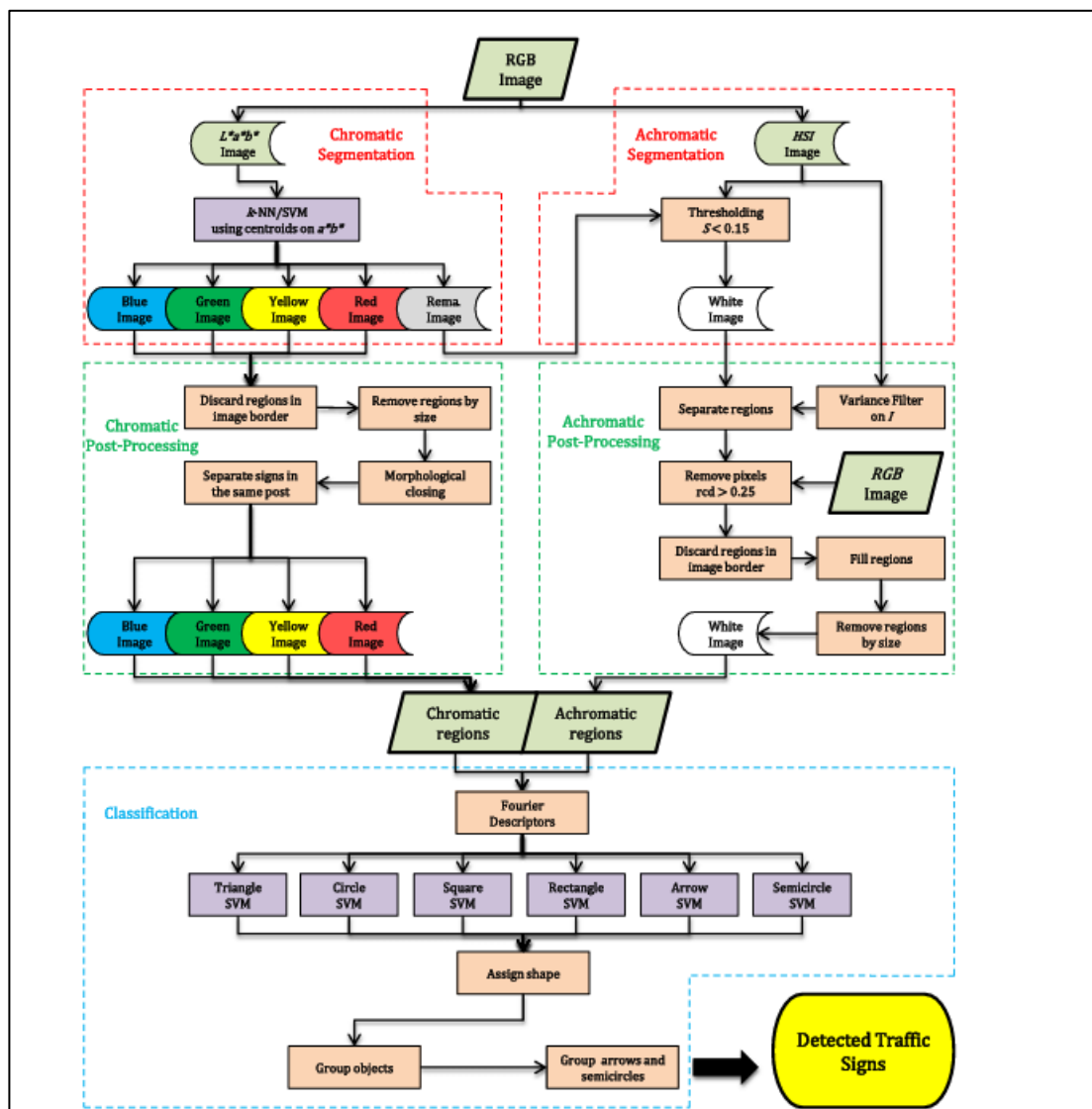


Figure 2.8 Architecture of Statistical Learning Methods

The system provides high classification accuracy (98.29%) and maintains low computational cost. Its lightweight nature makes it highly suitable for embedded platforms or lower-end processing units in smart vehicles or IoT devices.

Nevertheless, the performance is sensitive to lighting changes, affecting the accuracy of color-based segmentation. Signs that are worn out or blocked may not be effectively detected, and the method lacks robustness in real-world driving conditions.

2.2.2.10 Mexican Traffic Sign Detection and Classification Using Deep Learning

This research uses a two-stage pipeline involving Faster R-CNN for detection and a CNN classifier fine-tuned on a Mexican traffic sign dataset. The authors also apply data augmentation to improve the generalization of the model. (Rodríguez *et al.*, 2022)

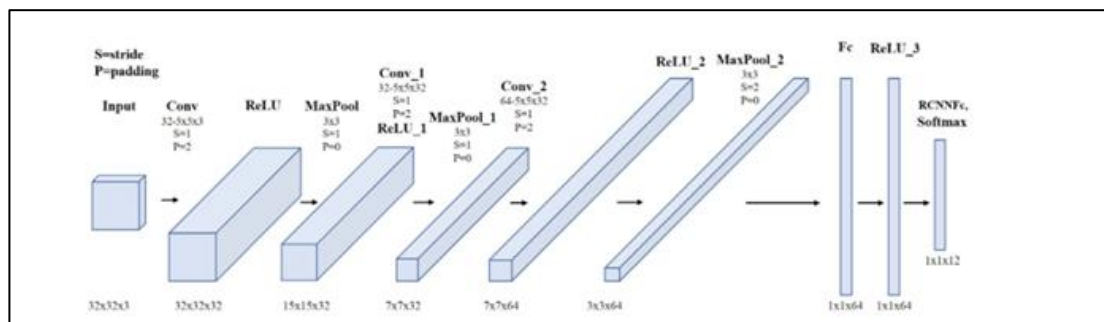


Figure 2.9 Architecture of R-CNN

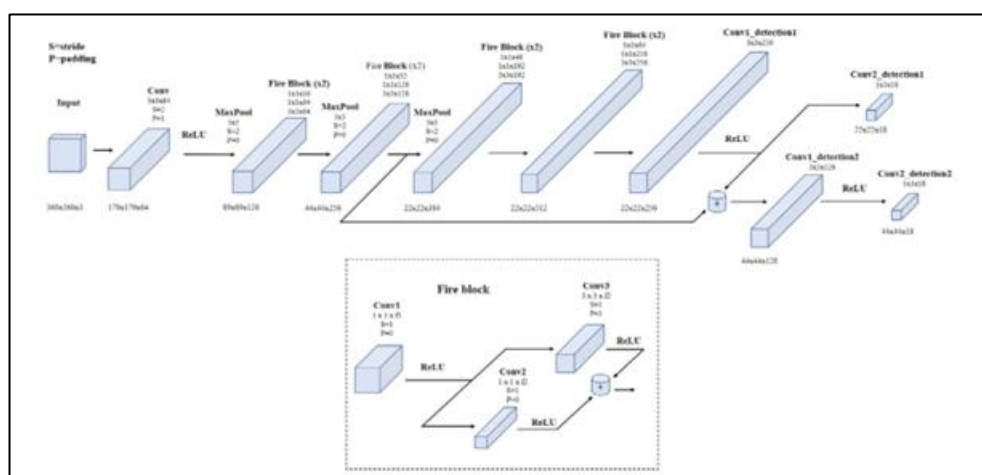


Figure 2.10 Architecture of YOLOv3

The system achieves high accuracy in both detection and classification on real-world road environments. Its use of data augmentation improves robustness to lighting and perspective variations, and the model demonstrates effectiveness in practical deployment conditions.

However, the system has high computational requirements for training and inference. It also relies on a dataset that is specific to Mexican signage, which may limit its adaptability to other countries without extensive retraining.

2.2.3 Review of Existing Traffic Sign Detection Systems

This section presents a critical review of six existing commercial or applied traffic sign detection systems. Each system is discussed in terms of its implementation, strengths, and limitations, with a focus on their relevance to road safety, driver assistance, and intelligent transportation.

2.2.3.1 NEOPLAN Traffic Sign Detection

The NEOPLAN Traffic Sign Detection system uses a camera to detect traffic signs and display them to the driver through the dashboard interface. It also provides visual and acoustic warnings when violations occur, such as overspeeding or ignoring regulatory signs. This system is typically integrated into commercial vehicles to improve road compliance.

The system enhances driver awareness by supplying real-time sign recognition and reduces manual workload, especially on unfamiliar routes. Automated detection helps drivers quickly recognize crucial signs and react appropriately, making it an effective driver assistance tool.

However, the system has limitations in low visibility environments. Its accuracy drops under poor lighting or adverse weather conditions. Furthermore, it may struggle to differentiate between visually similar signs without using higher-resolution imaging and advanced classifiers.

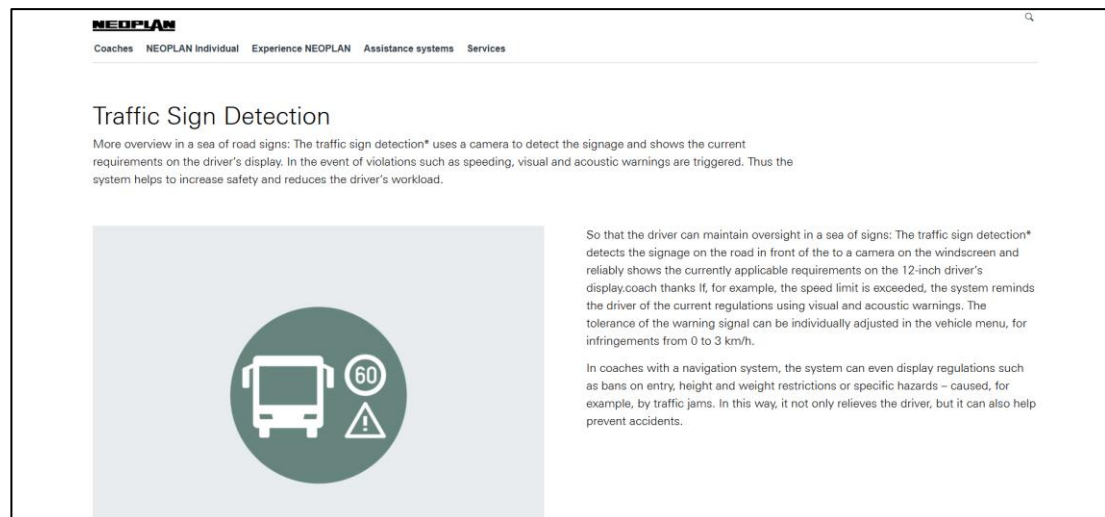


Figure 2.11 Dashboard Interface of NEOPLAN Traffic Sign Detection

2.2.3.2 Mobileye by Intel

Mobileye is an advanced ADAS solution developed by Intel that incorporates deep learning and edge AI for real-time detection of traffic elements such as road signs, pedestrians, lane markings, and nearby vehicles. It is equipped with CNNs trained on large-scale datasets and supports sensor fusion with LiDAR and radar for accurate perception.

The system operates without relying on cloud connectivity, offering fast alerts through edge processing. Its robustness in detecting multiple types of road elements makes it ideal for autonomous driving and ADAS integration in modern vehicles.

Despite these strengths, Mobileye models trained on global datasets may underperform in regions like Malaysia due to unique or less frequent signage. In addition, the system requires specialized cameras and certified hardware, which increases cost and limits accessibility for general consumer use.

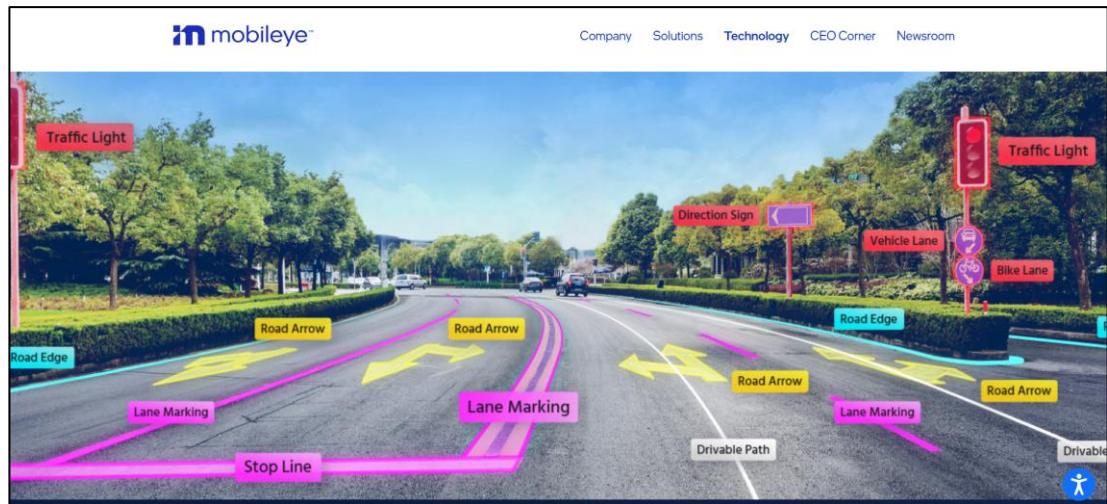


Figure 2.12 Dashboard Interface of Mobileye

2.2.3.3 MAN Traffic Sign Detection

The MAN Traffic Sign Detection system is a camera-based platform that detects road signs such as speed limits and other regulatory indicators. It alerts drivers with both visual and audio signals, which are displayed on the vehicle's onboard display system.

It is effective in preventing speeding violations by providing early warnings to the driver. Its seamless integration with the vehicle display ensures minimal distraction and consistent driver support during travel.

However, the system can face challenges in detecting traffic signs in crowded urban environments with many overlapping signs. Additionally, its efficiency is reduced in scenarios where signage is unclear, obstructed, or placed in non-standard locations.

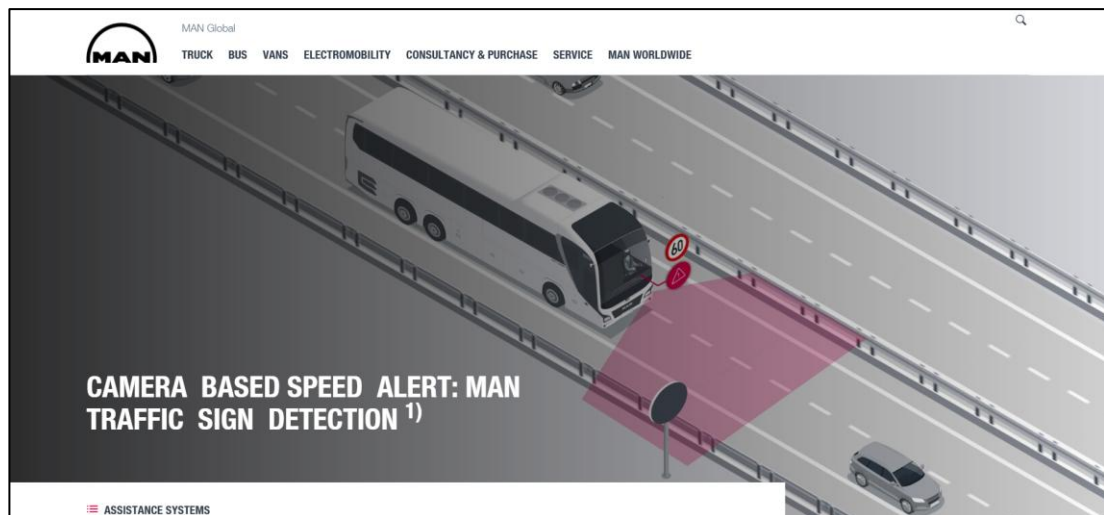


Figure 2.13 Dashboard Interface of MAN Traffic Sign Detection

2.2.3.4 ImageVision.ai Traffic Sign Detection

ImageVision.ai offers a computer vision-based system tailored to help traffic authorities monitor and maintain road signage. The system recognizes standard safety signs and identifies damaged or missing signs that require maintenance.

It plays a vital role in proactive road safety by ensuring that traffic signs are present, clear, and compliant. This enhances public safety by enabling authorities to act quickly on damaged infrastructure, potentially preventing accidents.

The system is highly reliant on image quality. If the signs are faded, blocked, or captured under poor lighting, detection accuracy drops. Furthermore, it is not designed for real-time in-vehicle use, making it more suitable for back-end analytics than driver assistance.



Figure 2.14 Dashboard Interface of ImageVision.ai Traffic Sign Detection

2.2.3.5 iSpeed – Traffic Sign Recognition

iSpeed is a Windows 10 application that transforms a standard device into a driving assistant by detecting speed limit signs using the device's camera. Upon detecting a sign, the system provides an audible alert to the driver.

Its biggest advantage is affordability. It uses existing hardware, and the auditory feedback enables drivers to keep their eyes on the road. This makes it an accessible option for budget-conscious drivers or early ADAS adopters.

However, it can only recognize speed limit signs within a red circle, limiting its applicability. Its performance is also compromised under low lighting or when signs are partially blocked from the camera view.

2.2.3.6 Signavi – Japanese Road Sign Recognition

Signavi is a real-time road sign detection app designed for Japanese roads. It uses a smartphone camera to recognize road signs and provides immediate voice feedback in multiple languages such as Japanese, English, Chinese, and Korean.

It is particularly beneficial for foreign tourists and non-native speakers driving in Japan. The system's ability to offer multilingual audio feedback enhances driving safety for international users and helps in sign comprehension.

However, the system is limited in scope as it only supports 11 core traffic signs including speed limits (30, 40, 50 km/h), stop signs, and no overtaking. This restricts its usefulness in more diverse or complex driving environments.

2.2.4 Comparison Between Existing Systems and the Proposed System

The **NEOPLAN Traffic Sign Detection** system offers visual and acoustic warnings to alert drivers about detected traffic signs. It integrates well into dashboard displays and reduces driver workload through automation. However, NEOPLAN's performance deteriorates under poor lighting or weather conditions and struggles with signs that are similar in appearance. In contrast, the proposed system is built using YOLO-based deep learning tailored specifically for Malaysian signs. It improves detection accuracy under varied environments and supports both real-time feedback and detection logs, offering a level of post-drive analysis that NEOPLAN lacks.

Mobileye by Intel is a powerful ADAS platform that uses advanced CNNs and supports sensor fusion with LiDAR and radar. It provides real-time alerts and operates on edge devices, which eliminates latency from cloud processing. However, Mobileye is trained on global datasets and may perform suboptimally with Malaysian-specific signs. It also requires proprietary hardware, making it costly and less accessible. The proposed system, in comparison, is designed for affordability and localization, trained on Malaysian datasets, and does not require expensive hardware—making it deployable on ordinary dashcams or mobile devices.

The **MAN Traffic Sign Detection** system integrates with the vehicle's dashboard to alert drivers about speed limits and restrictions. It helps prevent speeding and regulatory violations but faces difficulties in dense urban settings with overlapping or poorly placed signs. The proposed system improves upon this by employing a more adaptable deep learning model that can handle crowded or cluttered environments

through robust feature extraction. Additionally, it supports multi-class detection, enabling recognition beyond just speed limits—something MAN lacks.

ImageVision.ai focuses on detecting damaged or missing traffic signs for maintenance purposes. While it plays a crucial role in infrastructure management and public safety, it is not intended for in-vehicle use or real-time driver assistance. In contrast, the proposed system operates within the vehicle, providing both visual/audio alerts and data logging, which benefits both the driver and authorities. Its dual-purpose design bridges the gap between active driving support and post-drive evaluation.

iSpeed – Traffic Sign Recognition is a Windows-based application that provides audible alerts when it detects speed limit signs. Its biggest strength is its affordability, as it uses existing devices. However, it only supports speed limit signs within red circles and suffers under poor lighting or occlusion. The proposed system, while also lightweight, supports a broader range of sign classes and includes advanced object detection techniques that improve recognition accuracy even under low-visibility conditions, thereby offering greater versatility than iSpeed.

Signavi is a multilingual mobile app used for detecting a small subset of Japanese traffic signs. It is particularly useful for tourists and non-native speakers, offering voice alerts in various languages. However, the app is restricted to just 11 predefined signs, limiting its usefulness in broader driving contexts. The proposed system, by contrast, is designed for scalability and adaptability, capable of recognizing many classes of Malaysian traffic signs. Its modular structure also supports expansion to more sign types in the future, and its logging capability provides added value for enforcement and analytics.

After reviewing all the six existing systems—NEOPLAN, Mobileye, MAN TSD, ImageVision.ai, iSpeed, and Signavi—the key takeaway is that while each system offers useful features like in-dash integration, real-time alerts, and infrastructure monitoring, they often suffer from limitations in adaptability, local sign recognition, cost, or versatility. Most systems are either designed for broader global use without localization or rely on proprietary hardware that restricts accessibility. In contrast, my proposed system stands out by focusing on Malaysian-specific traffic sign detection using a cost-effective, flexible, and open-source approach. It not only detects

and classifies a wide range of sign types (including localized multilingual signs), but also offers logging, post-drive analysis, and potential future integration with enforcement systems or ADAS platforms.

2.2.5 Techniques

The following are the techniques and methods that are used to build and might be applicable to implement the traffic sign detection system:

2.2.5.1 Computer Vision and Deep Learning

Computer vision is a vital field of artificial intelligence that allows machines to interpret and understand the visual world through the processing of images and video. For this project, computer vision is used to detect and recognize Malaysian traffic signs in real-time from live dashcam or webcam video feeds. The system identifies different sign classes — such as speed limits, warnings, prohibitions, and informational signs — and issues visual or audio alerts accordingly.

The core of this system uses **Convolutional Neural Networks (CNNs)**, which are highly effective for image recognition tasks. CNNs have been widely adopted in traffic sign recognition tasks for their high classification accuracy (Sermanet & LeCun, 2011). CNNs automatically learn to identify patterns such as shapes, colors, and borders in traffic signs. A CNN classifier was trained to recognize different speed limit signs (30–110 km/h) using a dedicated dataset. The CNN model achieved 94% accuracy, proving effective for multi-class classification based on cropped images.

To handle the detection task, the system uses the **YOLO (You Only Look Once)** family of models. YOLO is a real-time object detection algorithm known for its speed and efficiency. Unlike traditional two-stage detectors that first generate region proposals and then classify them, YOLO operates as a single-stage detector. It divides the image into a grid and simultaneously predicts bounding boxes and class probabilities directly from full images in one evaluation. This makes it particularly suitable for time-sensitive applications like autonomous driving, where speed and accuracy are both crucial.

A more powerful object detection model, **YOLOv8 (You Only Look Once version 8)**, was used to detect traffic signs from full video frames. YOLOv8 is the latest and most advanced model from the YOLO family, offering real-time detection speed with high accuracy and supporting object detection, segmentation, and tracking tasks (Hussain, 2023). It supports multiple vision tasks such as object detection, segmentation, and tracking. The YOLOv8 model in this project was trained on a custom Malaysian traffic sign dataset, achieving 88.7% mAP@50, making it suitable for deployment in intelligent vehicles or mobile dashboard systems.

Several of the reviewed research papers in section 2.2.2 also support the use of these techniques. For example, Jiang et al. (2024) proposed YOLO-CCA, which enhances YOLOv7 with transformer-based context modules for better sign detection, while Ahmed et al. (2020) applied CNN-based classification in challenging weather conditions. These findings validate the decision to use a combination of YOLO for detection and CNN for classification in this project, ensuring both efficiency and robustness in real-world driving environments.



Figure 2.15 Example of real-time traffic sign detection output using YOLOv8.

2.2.5.2 Real-Time Video Processing

The project incorporates real-time video capture using either a USB webcam or dashcam, which feeds live video frames into the YOLOv8 detection model for

continuous processing. Additionally, the system is currently implemented to support uploaded video input, allowing detection to be performed on pre-recorded driving footage. This flexibility is especially useful for testing, demonstrations, and post-drive analysis. Whether the source is a live feed or uploaded clip, the video is processed frame-by-frame using OpenCV, which manages frame capture, resizing, and overlaying of detection outputs.

Each detected traffic sign is labeled with a bounding box, class label, and confidence score, which are rendered in real-time on the video output. Studies have shown that YOLO-based architectures, such as Attention-YOLOv4, are highly effective for real-time traffic sign detection with high accuracy (Li et al., 2022). Furthermore, the system includes a logging module that stores detection information — such as the timestamp of detection, and optionally, the detected frame itself. This log enables further review by drivers, researchers, or law enforcement personnel, supporting both real-time awareness and retrospective traffic compliance evaluation.

2.2.5.3 AI Integration in Smart Driving Systems

The integration of this detection system with Advanced Driver Assistance Systems (ADAS) presents a strong case for expanding its functionality. The modular design of the system allows its use as a standalone application or a plug-in module in smart vehicles. By issuing real-time alerts and recording interactions with traffic signs, the system contributes to both proactive safety (by informing the driver) and reactive review (through log analysis). This aligns with the current trend of integrating machine learning into ADAS components to reduce traffic accidents and enhance driving safety (Moujahid et al., 2021).

2.3 Project Methodology

2.3.1 Operational Framework – CRISP-DM

This project follows the **CRISP-DM (Cross-Industry Standard Process for Data Mining)** methodology, which provides a systematic approach for developing

data-driven systems. The six main phases of CRISP-DM were applied to guide the development of the *Real-Time Malaysian Traffic Sign Detection and Alert System*, integrating both object detection and classification using YOLOv8 and CNN models respectively.

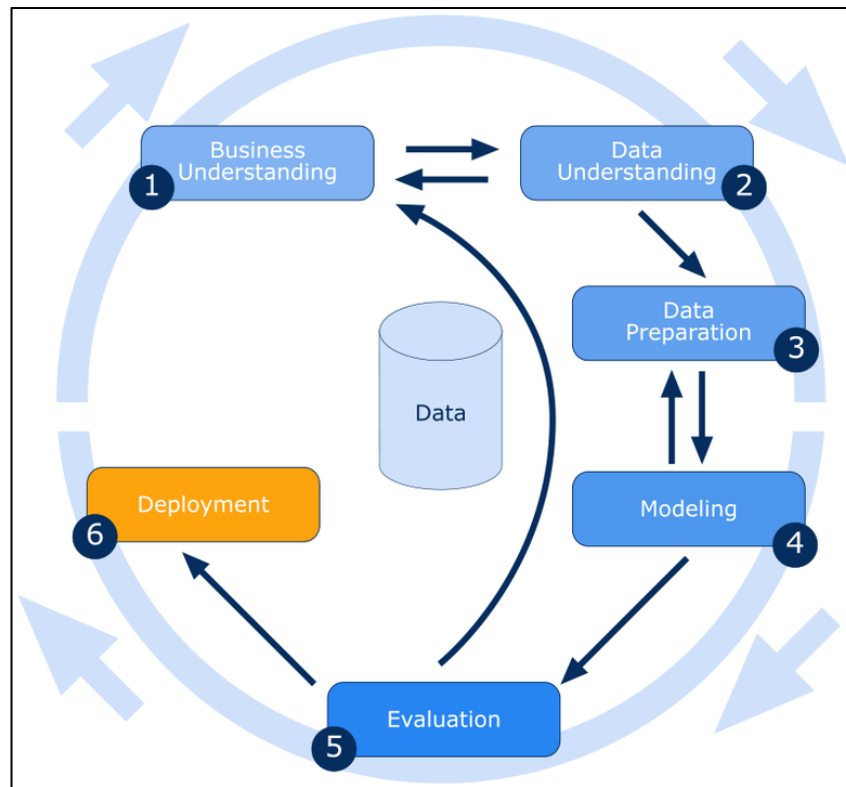


Figure 2.16 CRISP-DM Methodology

1. Business Understanding

The primary goal of this project is to develop a real-time traffic sign detection system focused on Malaysian road signs. The system is designed to improve driving safety by detecting and recognizing traffic signs from live video input and generating visual/audio alerts. Additionally, it logs detection events for further review by drivers or traffic enforcement agencies. Unlike global ADAS solutions, this system emphasizes local adaptability, lightweight deployment, and multi-stakeholder usability.

2. Data Understanding

Two major datasets were used:

- **YOLOv8-based detection dataset:** Included annotated Malaysian traffic signs across classes like “Speed Limit 30,” “Children Crossing,” “No Entry,” and more, prepared in YOLO format using Roboflow.
- **CNN classification dataset:** Focused specifically on classifying speed limit signs (e.g., 30, 50, 60, 70, 90, 110 km/h) into distinct speed categories.

Data exploration included analyzing class distribution, detecting imbalances, and reviewing image quality. Images were collected in varied lighting and road conditions to reflect real-world environments.

3. Data Preparation

i. For YOLOv8:

- Images were resized, and annotations were converted to YOLOv8 format.
- Augmentation techniques like rotation, brightness variation, and flipping were applied using Roboflow and Ultralytics tools.
- Dataset was split into **training (80%)**, **validation (10%)**, and **testing (10%)**.

ii. For CNN:

- Speed limit images were resized to uniform dimensions (e.g., 64×64).
- One-hot encoding was applied for label processing.
- The dataset was normalized to improve CNN convergence and generalization.

4. Modeling

Two models were trained:

i. **YOLOv8 Object Detection Model:**

- Trained using Ultralytics' YOLOv8 on the detection dataset.
- Hyperparameters like batch size, learning rate, and epochs were tuned.

- Training was conducted on Google Colab and saved using checkpoints (best.pt).
- ii. **CNN Speed Limit Classification Model:**
- A convolutional neural network was implemented using PyTorch.
 - The model was trained to classify cropped sign images into speed categories (30–110).
 - The final CNN weights were saved as best_speed_class_cnn.pt.

Both models achieved satisfactory accuracy and detection precision. YOLOv8 served as the primary real-time detection engine, while CNN acted as a secondary classifier for speed signs if cropped for further classification.

5. Evaluation

i. **YOLOv8:**

- Achieved a mean Average Precision (mAP@50) of approximately 88.7% across major traffic sign classes, and mAP@50–95 of 67.2%, indicating strong detection accuracy suitable for real-time applications.
- Precision and recall were also measured using validation sets.
- Real-time test video showed successful detection with labeled boxes and confidence scores.

ii. **CNN Model:**

- CNN model achieved an accuracy of 94% in classifying speed limit signs, demonstrating reliable performance in distinguishing speed categories.
- Confusion matrix showed strong separation between speed categories, with minor confusion between visually similar signs (e.g., 60 vs 90).

This dual-model approach ensured high confidence in both detection and classification, especially for signs that require precise enforcement such as speed limits.

6. Deployment

The YOLOv8 model was deployed in a real-time Python application. A live video feed (or uploaded video) was processed frame-by-frame, and:

- Detected signs were drawn with bounding boxes and class labels.
- Alerts were displayed visually on the dashboard.
- Detected frames with timestamped logs were saved for later viewing.
- The CNN model can optionally be integrated as a classifier module for speed sign refinement if needed.

This modular architecture ensures the system can be embedded into ADAS dashboards or mobile applications, offering future compatibility with enforcement systems and smart vehicle platforms.

2.4 **Project Requirements**

This project involves developing a real-time traffic sign detection and alert system using deep learning. It requires a well-defined set of software, hardware, and support resources. The following subsections outline the key components necessary for implementing, training, and deploying the system effectively.

2.4.1 **Software Requirements**

The following software tools were used throughout this project for dataset management, model development, real-time processing, and UI design:

1. Visual Studio Code

A flexible code editor used for writing Python scripts, real-time display modules, and logging mechanisms. Its Git and debugging integrations streamlined iterative testing during development.

2. Jupyter Notebook

Essential for experimenting with data preprocessing, training YOLOv8 and CNN models, and visualizing training metrics. It allowed modular and annotated code development with visual feedback.

3. Google Colab

Provided free access to GPU hardware to accelerate training of deep learning models. Used for training the YOLOv8 object detector and CNN classifier efficiently.

4. Python 3.x

The primary programming language for the project, supporting TensorFlow, PyTorch, OpenCV, and Ultralytics YOLO libraries.

5. OpenCV

Used for real-time video frame capture, processing, and overlaying bounding boxes during detection.

6. Ultralytics YOLOv8 Library

Used to train and deploy the object detection model with real-time inference support.

7. PyTorch

Utilized to train the CNN model for speed limit sign classification with custom architecture.

2.4.2 Hardware Requirements

The table 2.2 outlines the hardware used and recommended to ensure smooth training and real-time deployment of the system:

Table 2-2 Hardware used and recommended

Component	Minimum	Recommended
<u>Processor</u>	2.3 GHz quad-core	2.3 GHz 8-core or better
<u>Memory</u>	8 GB RAM	16 GB RAM or more
<u>Graphics Card</u>	NVIDIA GTX 1050 Ti	NVIDIA RTX 2060 or higher
<u>Storage</u>	256 GB SSD	512 GB SSD or more
<u>Camera</u>	USB/Webcam or dashcam	Full HD USB camera

2.4.3 Other Requirements

- Malaysian Traffic Sign Dataset
Acquired from Roboflow, Kaggle and Malaysian Traffic Sign Dataset Article, including classes such as Speed Limit, Bumps, No Entry, and Expressway Signs.
- High-speed Internet Connection
Required for Google Colab training, dataset upload, and YOLO model downloading.
- GPU Access
Provided via Google Colab (NVIDIA Tesla T4 or P100) to accelerate deep learning model training.

2.4.4 Project Schedules and Milestones

This section presents the timeline and deliverables for the system development using project management practices. It outlines stages from research and dataset collection through deployment and report writing.

No	Task	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	Discussion of Title and Project with Supervisor														
2	Project Progress 1														
3	Report Writing Progress 1														
4	Project Progress 2 (System Development)														
5	Report Writing Progress 2														
6	Demonstration/Presentation with Supervisor and Evaluator														
7	Report Evaluation														

Figure 2.17 Gantt Chart

2.5 Summary

This chapter presented a comprehensive literature review and detailed project methodology focused on traffic sign detection and alert systems in the context of Malaysian road safety. The introduction emphasized the growing importance of real-time detection systems for improving driver awareness, supporting ADAS integration, and assisting law enforcement. The following sections critically reviewed related

academic research and commercial systems to identify current capabilities and limitations in the field.

The review of academic literature included various deep learning and machine learning techniques such as YOLOv7 with context augmentation, VSSA-NET with spatial attention, and classical approaches using SVM and HOG. Each technique was evaluated for its detection accuracy, computational efficiency, and suitability for real-time deployment. The review also covered localized traffic sign classification using Faster R-CNN and CNN models, with particular attention to Malaysian-relevant signage.

Commercial systems such as Mobileye, MAN, and NEOPLAN were analyzed to highlight their strengths, including dashboard integration and advanced sensor fusion, as well as their drawbacks like high costs, limited flexibility, or lack of localization. Compared to these, the proposed system aims to provide a lightweight, cost-effective, and Malaysian-focused detection platform using YOLOv8 and CNN, along with logging capabilities and potential enforcement integration.

The methodology section detailed the use of the CRISP-DM framework, starting from business understanding and data acquisition to preprocessing, model training, evaluation, and deployment. The YOLOv8 model was trained using annotated Malaysian traffic signs and achieved a mean Average Precision (mAP@50) of 88.7%. Meanwhile, the CNN model for speed sign classification achieved 94% accuracy. The system was then deployed in a real-time video processing setup, capable of displaying live detections, generating alerts, and logging sign data.

Project requirements were also specified, covering both software tools (such as Google Colab, PyTorch, and OpenCV) and hardware needs (including GPU-enabled systems and USB cameras). A structured timeline with well-defined milestones, aligned with the CRISP-DM process, ensured that the development was systematic and goal-driven. This timeline included phases for data preparation, model training, real-time integration, and final evaluation.

In conclusion, this chapter laid a strong foundation for the system's development by integrating theoretical research, practical experimentation, and

structured methodology. It demonstrated a well-organized and localized approach to solving traffic sign detection challenges in Malaysian driving environments and set the stage for the implementation and testing phases to follow in later chapters.

CHAPTER 3: REQUIREMENT ANALYSIS

3.1 Introduction

This chapter discusses the problem analysis and requirement specification of the proposed Real-Time Malaysian Traffic Sign Detection and Alert System. It outlines the current system scenario and presents a data flow diagram (DFD) that visualizes the interaction between components. Furthermore, this chapter defines the data inputs required, the system's functional and non-functional requirements, and the expected features to ensure successful implementation.

3.2 Problem Analysis

3.2.1 Current System Scenario

Currently, most vehicles rely on manual observation of traffic signs, which depends heavily on the driver's attention and visibility conditions. Drivers may miss signs due to fatigue, distractions, weather conditions, or poor signage placement. Although commercial solutions like Mobileye offer traffic sign detection, they are expensive, not tailored for Malaysian traffic environments, and require dedicated hardware.

The absence of a localized, affordable, and real-time system leads to unsafe driving behaviours and reduced road safety. This project aims to bridge this gap by developing a deep learning-based detection system that can identify Malaysian road signs in real time using camera input from a USB webcam, dashcam, or uploaded video. The system will provide visual/audio alerts and log detections for further review.

In addition, current traffic enforcement and monitoring by JKR still rely on manual methods such as physical inspections, speed traps, or post-incident dashcam reviews. These approaches are time-consuming and resource-intensive, and they do not provide continuous monitoring. By integrating a real-time logging mechanism, this project enables JKR to access timestamped evidence of traffic sign encounters, analyze

driver compliance, and identify accident-prone zones. This dual focus on both drivers and authorities makes the system practical for improving road safety and governance.

3.2.2 Data Flow Diagram (DFD)

The Data Flow Diagram (DFD) Level 0 illustrates the overall workflow of the Real-Time Traffic Sign Detection System for both drivers and JKR reviewers. It highlights the flow of data between users, system components, storage, and dashboards, focusing on how video input is processed, detections are logged, and how JKR accesses reports.

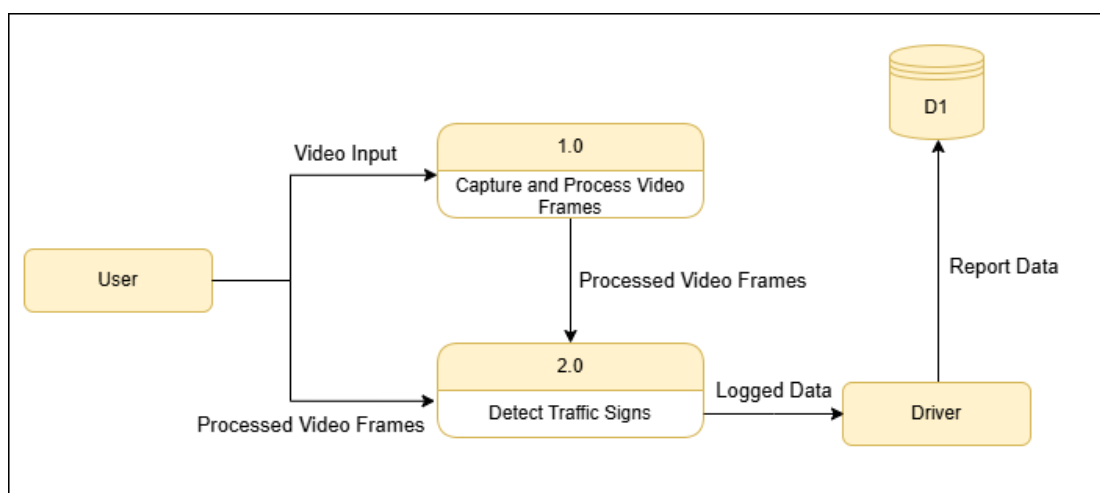


Figure 3.1 Data Flow Diagram (DFD) Level 0

Main Process: Traffic Sign Detection System

This main process begins with the user supplying a video (via webcam or uploaded file), which is processed to detect traffic signs in real-time. The system displays the results, generates alerts, and logs the detection for future reference.

External Entities

1. Driver

In Provides video input and receives real-time alerts and a basic log view.

2. JKR Reviewer

Accesses the JKR Dashboard to query, filter, and export detection logs (e.g., by date, sign type, location if available) for audits, hotspot analysis, and reporting.

System Processes

1. Capture and Process Video Frames (Process 1.0):

- Accepts the input video and extracts frames.
- Prepares frames for detection.

2. Detect Traffic Signs (Process 2.0):

- YOLOv8 model is applied to each frame.
- Detected traffic signs are logged with a timestamp and frame image.
- Results are passed to the driver for display and review.

Data Store

1. D1 – Detection Log:

- Stores detection results include timestamp and annotated image.
- Can be accessed for post-drive review or enforcement reporting.

3.2.3 Flowchart

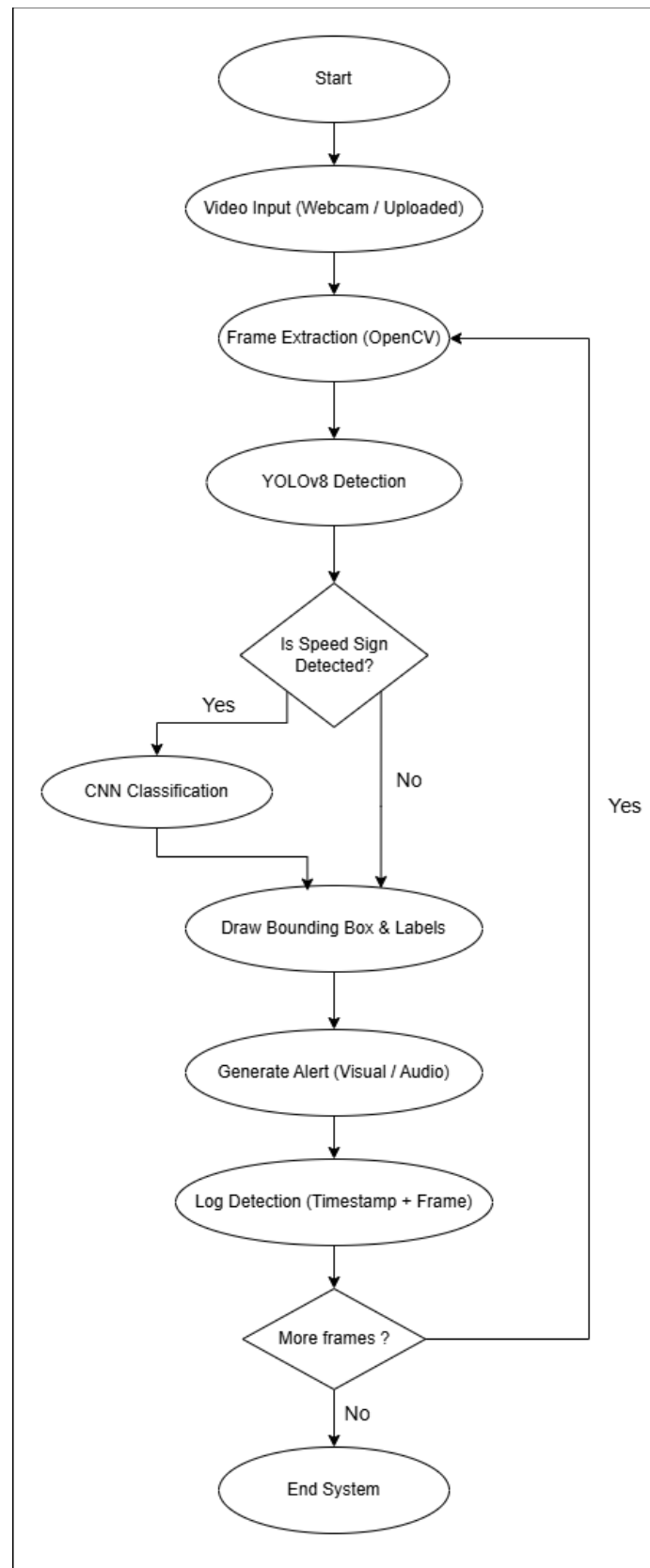


Figure 3.2 Flowchart of Real-Time Malaysian Traffic Sign Detection and Alert System

Step-by-Step Description

1. Video Input (Webcam / Uploaded)

- The system starts by accepting input either from a live webcam/dashcam feed or a pre-recorded video file uploaded by the user.

2. Frame Extraction (OpenCV)

- Using OpenCV, the video is processed frame-by-frame, extracting each frame to be passed into the detection pipeline.

3. YOLOv8 Detection

- The extracted frame is sent to the YOLOv8 object detection model, which identifies traffic signs by drawing bounding boxes and assigning class labels (e.g., No Entry, Speed Limit, Road Work).

4. Is Speed Sign Detected?

- A decision point checks if the detected sign is a speed limit sign.
- If yes, the detected region (bounding box) is cropped and sent to a CNN classifier for more precise classification (e.g., 30, 50, 70, 90 km/h).
- If not, the system proceeds directly to display the detection results.

5. CNN Classification (Only for Speed Signs)

- The cropped sign image is passed through a CNN model that classifies the exact speed limit value.

6. Draw Bounding Box & Labels

- Once classification is complete (or skipped), the system overlays detection labels, bounding boxes, and confidence scores onto the frame.

7. Generate Alert (Visual & Audio)

- If the detected sign is critical (e.g., No Entry, Over Speed, School Zone), a real-time alert (e.g., sound and visual flash) is generated to notify the driver.

8. Log Detection (Timestamp + Frame)

- Every detection is logged with details such as:
 - Timestamp
 - The detected frame image
- These logs can be reviewed later by drivers or enforcement.

9. More Frames?

- If the video has more frames to process, the system loops back to Frame Extraction and repeats.
- If no frames are left, the system proceeds to the final step.

10. End System

- The detection loop ends, and all logs are saved for post-review or report generation.

3.3 Requirement Analysis

3.3.1 Data Requirement

To build a functional and accurate Real-Time Traffic Sign Detection and Classification System tailored for Malaysian roads, several specific types of data are required. First and foremost, the system relies on a large collection of annotated traffic sign images for training the YOLOv8 object detection model. These images must include a variety of Malaysian traffic signs such as speed limit signs (e.g., Speed limit 20 to 110), warning signs (e.g., Bumps, Children), prohibitory signs (e.g., No Entry, No Parking), and other regulatory or directional signs (e.g., Chevron, Expressway). YOLO-based models have proven effective for real-time detection of small traffic signs in challenging environments (Zhang et al., 2020).

In addition to the detection dataset, a separate dataset of cropped speed limit signs is used to train a Convolutional Neural Network (CNN) model, specifically a fine-tuned ResNet18. This classifier is responsible for accurately recognizing the specific speed limit value (such as 30, 50, 70, or 90 km/h) after YOLO detects a generic

“Speed limit” label. The cropped images are resized to 64×64 pixels and normalized using standard preprocessing techniques before being passed into the CNN. This two-stage approach—YOLO for detection and CNN for classification—ensures both efficiency and precision in identifying traffic signs.

Furthermore, the system requires real-time or recorded video input as the primary source of frames. Users may either upload a pre-recorded video file in formats such as .mp4, .avi, or .mov, or allow the system to capture live feed using a USB webcam or dashcam. Finally, the system incorporates a detection logging mechanism. Every time a traffic sign is detected, the system records the detection time (timestamp), the identified sign type, and optionally saves the annotated image frame. These records are stored in structured folders based on the date and can be used for future analysis, law enforcement review, or performance monitoring.

3.3.2 Functional Requirement

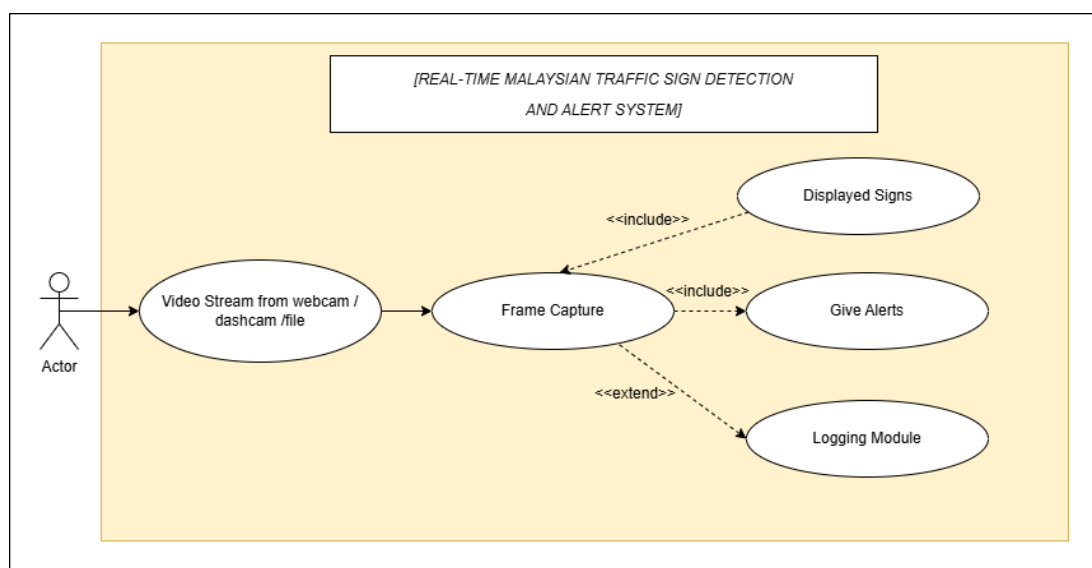


Figure 3.3 Use Case Diagram – Real-Time Malaysian Traffic Sign Detection & Alert System

Use Case 1: Start Detection System

- **Actor:** User (Driver/Admin)
- **Description:**
The user uploads a video or enables the live webcam/dashcam input to start the detection process.

- **Pre-conditions:**
 - The system is running.
 - Video or webcam is accessible.
- **Post-conditions:**
 - The video is processed frame by frame.
 - Detection begins automatically.
- **Main Flows:**
 - User uploads a video or activates the camera.
 - System receives the input stream.
 - Frame-by-frame processing starts.
- **Alternative Flows:**
 - **AF1.1:** If the uploaded file format is unsupported, the system shows an error and prompts for re-upload.
 - **AF1.2:** If webcam access fails, the system suggests checking the device or switching to upload mode.

Use Case 2: Detect and Classify Traffic Signs

- **Actor:** System
- **Description:**

Frames are processed using YOLOv8 for detection. If a speed sign is found, it is passed to the CNN model for classification.
- **Pre-conditions:**
 - Video frames are available.
 - YOLOv8 and CNN models are loaded.
- **Post-conditions:**
 - Detected signs are labeled.
 - Critical signs are sent to the alert module.

- **Main Flow:**
 - YOLOv8 model detects signs from each frame.
 - Speed limit signs are passed to the CNN for fine-grained classification.
 - Bounding boxes and confidence values are generated.
- **Alternative Flows:**
 - **AF2.1:** If no sign is detected in the frame, the system silently skips to the next frame.
 - **AF2.2:** If a corrupted frame is encountered, it is ignored and processing continues.

Use Case 3: Generate Alerts

- **Actor:** System
- **Description:**
Alerts are triggered when critical signs are detected
- **Pre-conditions:**
 - Detection of a critical traffic sign.
 - Audio & visual system is functional.
- **Post-conditions:**
 - An alert is displayed and/or played to the user.
- **Main Flow:**
 - System checks if detected label matches critical alert list.
 - If match found, audio or visual alert is triggered.
 - Alerts are overlaid on the frame in real time.
- **Alternative Flows:**
 - **AF3.1:** If multiple critical signs are found, one with highest priority is alerted first.
 - **AF3.2:** If the system cannot play audio, only the visual alert is shown.

Use Case 4: View Detection Logs

- **Actor:** User (Driver/Reviewer)
- **Description:**
Allows the actor to view stored detection logs (frames + timestamps) for review, reporting, or enforcement. Drivers access logs through the Driver Dashboard, while JKR reviewers access logs through the JKR Dashboard for compliance monitoring and safety audits.
- **Pre-conditions:**
 - Detections were previously logged.
 - Logs are saved in system storage (organized by date).
- **Post-conditions:**
 - The user can view or download detection results.
 - JKR reviewers can export detection logs in CSV/XLSX/PDF format directly from the JKR Dashboard.
- **Main Flow:**
 - User selects a specific date from the dashboard.
 - System fetches detection images with timestamps.
 - Each image can be opened in full view.
- **Alternative Flows:**
 - **AF4.1:** If no detections exist for selected date, a message is shown: "No logs available."
 - **AF4.2:** If an image is missing/corrupted, it is skipped and others are shown.
 - **AF4.3:** If accessed by JKR, the reviewer can choose to export the logs as a report for official use.

3.3.3 Non-Functional Requirement

- **Performance:** The system should process at least 1 frame per second in real-time mode.

- **Accuracy:** The detection model should maintain a minimum of 85% precision for key sign classes.
- **Usability:** The interface should be simple, responsive, and user-friendly.
- **Scalability:** The system should allow future integration with ADAS platforms or law enforcement dashboards.
- **Portability:** The model should run on any PC with at least 8GB RAM and a standard GPU or Google Colab.
- **Maintainability:** The system code should follow modular programming practices to support future updates.
- **Accountability:** Logs should be tamper-proof so JKR can trust the evidence.
- **Interoperability:** Logs are exportable in standard formats (CSV/XLSX/PDF) for integration into JKR's systems.

3.4 Summary

This chapter comprehensively analysed the system requirements and functional design of the proposed Real-Time Traffic Sign Detection and Classification System tailored for Malaysian roads. It began by identifying the data requirements, including annotated traffic sign images for YOLOv8 training and cropped speed limit signs for CNN-based classification using a fine-tuned ResNet18. The system supports both real-time webcam/dashcam input and uploaded video formats, with OpenCV used for frame handling. A detection logging mechanism records timestamps and saves annotated frames for future review. Functional and non-functional requirements were clearly defined to ensure performance, usability, and reliability. The Data Flow Diagram (DFD) illustrated the high-level workflow of video processing, detection, alerting, and log management. Additionally, the system's operation was visualised through a process flowchart, and a use case diagram captured all user interactions, including video upload, detection monitoring, alert handling, and log viewing. Altogether, this chapter lays the groundwork for implementing a robust, real-time, and multi-stakeholder traffic sign detection solution designed specifically for Malaysian traffic conditions.

CHAPTER 4: DESIGN

4.1 Introduction

This chapter presents the detailed design of the Real-Time Malaysian Traffic Sign Detection and Alert System. The system incorporates multiple components including user interface, backend logic, database architecture, and artificial intelligence models. Each subsection provides insights into the high-level architecture, system components, and visual representations to help understand how the system functions as a whole.

4.2 High-Level Design

4.2.1 System Architecture

The system is composed of three main modules: video input interface, AI-based detection and classification engine, and a web dashboard for visualization and alerting.

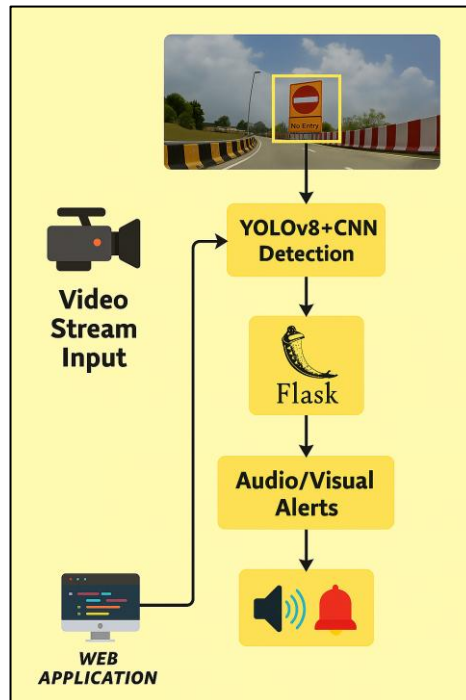


Figure 4.1 System Architecture Diagram

The updated system architecture diagram visually represents the core components and their interaction in the traffic sign detection and alert system:

- **Video Stream Input:** The process begins with video input, either from a dashcam or uploaded footage. This raw video stream provides the frames needed for detection.
- **YOLOv8 + CNN Detection:** The system processes each video frame through a hybrid AI engine. YOLOv8 handles object detection, identifying the position of traffic signs, while a CNN (ResNet-18) is used for classifying specific types like speed limits.
- **Flask Backend:** The detection results are then sent to a Flask-based backend. This module acts as the processing hub — managing data flow, preparing the outputs, and coordinating alerts.
- **Audio & Visual Alerts:** For critical detections (e.g., Road Work, Speed Limit), the backend triggers real-time alerts — both visual highlights and audio prompts to notify users.
- **Web Application:** Simultaneously, the data is sent to a web dashboard where users can view live detection outputs, logs of previously detected frames, and interact with the system in real-time.

4.2.2 User Interface Design

4.2.2.1 Input Design

The system adopts a simple, upload-only input design where users submit pre-recorded road scene videos for detection. This approach streamlines interaction and avoids the complexity of live camera integration at this stage of development.

The video upload interface is located prominently on the main dashboard of the web application. Users are prompted to select a file using the “**Choose File**” button. The system accepts only .mp4 and .avi formats, ensuring compatibility with standard video outputs from dashcams and smartphones.



Figure 4.2 Upload Video Interface

Once a file is selected:

- The filename appears in the input field (e.g., 1.mp4)
- Users click “**Upload & Detect**” to initiate the detection process

To prevent invalid inputs, the system includes **client-side validation** that:

- Checks for non-empty file selection
- Verifies the file extension is supported
- Prevents unsupported or corrupt uploads

This design ensures that only valid, processable videos reach the detection pipeline, improving performance reliability. The simplicity also enables users to upload and test videos quickly, whether for development, simulation, or demonstration purposes.

4.2.2.2 Output Design

The output design of the system focuses on delivering real-time and post-processing feedback to users in a visually intuitive manner. It includes two key components: **live detection output** during video playback, and a **detection log view** that stores and displays past detection results.

1. Live Detection Output

When a user uploads a video and initiates detection, the system processes each frame sequentially and overlays the detection results directly onto the video feed. These results include:

- **Bounding Boxes:** Rectangular markers are drawn around detected traffic signs.
- **Class Labels with Confidence Score:** Each bounding box includes a descriptive label (e.g., *Speed Limit 50*, *No Entry*) along with the model's confidence score (e.g., *0.92*) displayed beside it. These labels are rendered in a clear font with a consistent yellow background, ensuring visibility against various video environments.
- **Frame Synchronization:** The detections are synchronized with playback, ensuring the annotations match the sign's position in time.

This real-time visual feedback allows users to understand exactly where and when each traffic sign appears within the video.

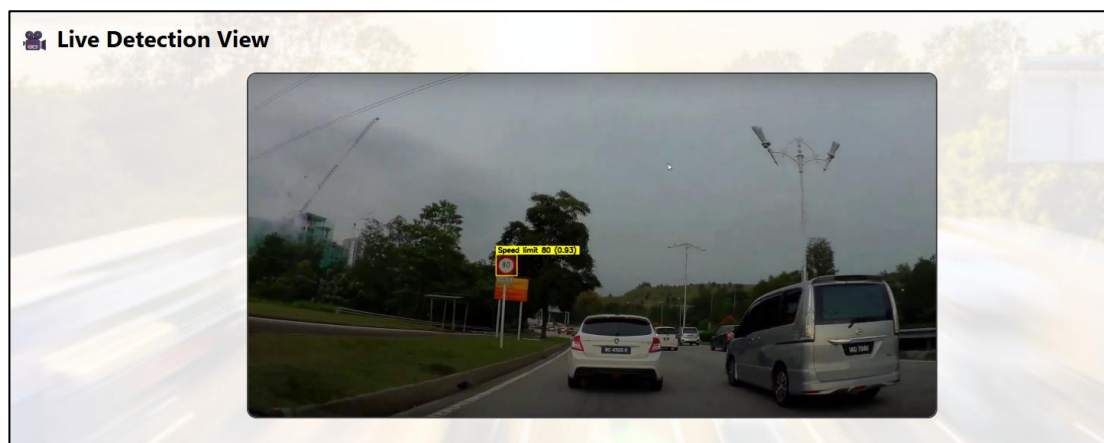


Figure 4.3 Live Detection Output

In addition to the bounding boxes and labels, the system incorporates both **visual and audio alerts** to highlight the detection of **critical traffic signs**. When a sign such as *No Entry* or *Speed Limit 110* is detected, a flashing or emphasized indicator appears on the screen — usually below or beside the video player — and an audio notification is played simultaneously to alert the user.

While the **visual alert** is clearly visible in the interface, the **audio alert** is triggered in the background using pre-recorded voice clips corresponding to each critical sign. These audio cues provide an additional layer of real-time feedback, especially useful in dynamic driving environments or for accessibility purposes.

The visual alert remains on screen for a brief period and is time-synced with the detection frame to ensure clarity and relevance. Due to the nature of audio output, it is not directly shown in the screenshot but functions alongside the visual prompt.

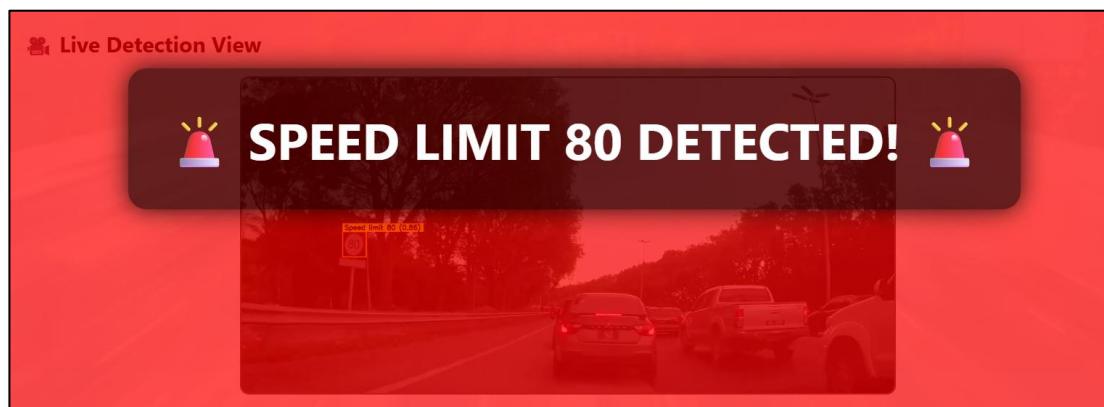


Figure 4.4 Visual Alert Example

A flashing red overlay and alert message are triggered when a critical traffic sign (e.g., Speed Limit 80) is detected. This visual cue appears in sync with the detection frame and is accompanied by an audio alert in the live system.

2. Detection Log View

In addition to real-time overlay during playback, the system provides a comprehensive log view that stores and displays frames where traffic signs were detected. This feature enhances usability by allowing users to review past detections, filter by date, and inspect frame-level details independently of the video.

The detection log is divided into two views:

i. Today's Detection View

Immediately after a video is processed, detection results are shown in the "**Today's Detections**" section. This view presents:

- All frames with detected traffic signs from the current session
- Organized in a grid layout
- Each image includes bounding boxes and labels overlaid directly on the frame

This helps users confirm what the system detected in the current video before switching to historical logs.



Figure 4.5 Today's Detection Grid

ii. Historical Detection Log by Date

Users can browse logs from previous sessions using the “**View Detection Log by Date**” feature. Selecting a date (e.g., 2025-06-12) displays:

- A folder-style log view
- Each detection displayed as a card with:
 - Detection label thumbnail
 - Exact timestamp (e.g., 12:43:06:029574)
- Detections sorted in reverse chronological order (latest first)

This archive enables persistent access to logs for auditing, verification, or demonstration purposes.



Figure 4.6 Detection Log for 2025-06-12

iii. Detailed Log Image Viewer

Clicking on any detection thumbnail enlarges the image in a lightbox-style view. This expanded view displays:

- The full frame
- All annotated signs with their bounding boxes and confidence scores
- Clear visibility for inspecting sign position and detection quality

This feature adds precision, allowing deeper inspection of detection confidence and alignment.



Figure 4.7 Detailed Log Frame View

4.2.2.3 JKR Dashboard

The JKR Dashboard is designed specifically for enforcement and road safety monitoring purposes. Unlike the driver dashboard, which focuses on live detection and alerts, the JKR Dashboard emphasizes data review, filtering, and analysis.

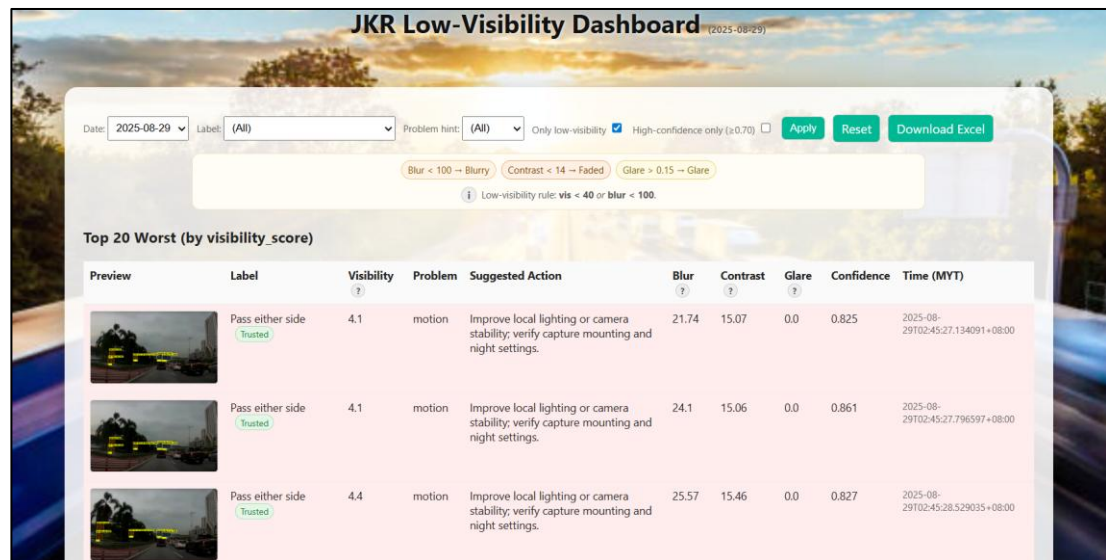


Figure 4.8 JKR Low-Visibility Dashboard

Key features include:

- **Date & Label Filters:** Reviewers can select a specific date and sign category to view detections.
- **Problem Hint Filter:** Allows reviewers to isolate detections with issues such as blur, low contrast, or glare.
- **Low-Visibility Filtering:** A toggle is provided to display only low-visibility cases, helping JKR to quickly identify problematic signs or road conditions.
- **Quality Metrics:** Each detection log includes blur, contrast, glare, and visibility scores. These indicators help JKR understand whether poor detection is due to environmental conditions or camera setup.
- **Suggested Actions:** The system provides recommended corrective measures (e.g., “Improve local lighting or camera stability”), which can assist JKR in road infrastructure audits and planning.

- **Preview and Confidence Level:** Each detection entry includes a preview image with bounding boxes, class label, confidence score, and timestamp in MYT.
- **Export to Excel:** Reviewers can export filtered logs into Excel format (CSV/XLSX) for compliance reports, enforcement checks, or long-term record keeping.

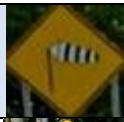
This dashboard transforms detection logs into actionable insights for JKR reviewers, enabling them to identify accident-prone zones, recurring visibility issues, and compliance trends without manually sifting through raw video footage.

4.2.3 Database Design

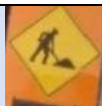
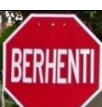
4.2.3.1 Dataset

Table 4-1 YOLOv8 Traffic Sign Class List with Labels, Images, and Descriptions

<u>YOLO Class ID</u>	<u>Class Label</u>	<u>Image</u>	<u>Description</u>
0	Bicycle lane		Designates a specific lane for bicycle traffic only.
1	Bumps		Warns of speed bumps or road humps ahead. Includes Bumps Ahead.
2	Bus stop		Indicates an area reserved for bus stopping or pickup.
3	Camera operation zone		Notifies drivers of areas monitored by traffic enforcement cameras.
4	Chevron -left-		Guides drivers through a sharp curve to the left.
5	Chevron -right-		Guides drivers through a sharp curve to the right.

6	Children		Warns drivers of a nearby school or children crossing area.
7	Construction sign		Alerts drivers to ongoing road construction ahead.
8	Cow nearby		Indicates a rural area where cows or livestock may be present on the road.
9	Crossroad		Warns of an upcoming intersection with roads crossing.
10	Crossroad on the left		Indicates a side road joining from the left.
11	Crossroad on the right		Indicates a side road joining from the right.
12	Crosswind area		Alerts to strong side winds that may affect vehicle stability.
13	Double Bend to Left Ahead		Warns of two consecutive bends, starting to the left.
14	Double Bend to Right Ahead		Warns of two consecutive bends, starting to the right.
15	Expressway sign		Marks the entrance or presence of an expressway. Merged from signs 1 & 2.
16	Flagman ahead		Indicates a flagman is ahead managing traffic, usually in construction zones.
17	Give way		Requires drivers to yield to oncoming traffic.
18	Height limit		Restricts access to vehicles taller than the specified height.

19	Horn Prohibited		Indicates an area where honking is not allowed.
20	Left Bend Ahead		Warns of a sharp bend to the left ahead.
21	Level crossing with gates ahead		Indicates a railway crossing protected by gates or barriers.
22	Motorcycles only		Denotes a path or lane reserved for motorcycles.
23	Narrow bridge		Warns of a bridge with restricted width for vehicles.
24	No Stopping		Prohibits vehicles from stopping at any time along that stretch.
25	No U-turns		U-turns are not permitted at this junction or section.
26	No entry		Entry is not allowed from the current direction.
27	No left turn		Vehicles are prohibited from making a left turn at the indicated spot.
28	No overtaking		Restricts vehicles from overtaking other vehicles in that zone.
29	No parking		Parking is not allowed at the location.
30	No right turn		Prohibits right turns from the current lane or road.
31	Obstruction		Indicates a physical obstruction or road blockage ahead.
32	Other dangers nearby		Warns of unspecified hazards in the area.
33	Parking area		Indicates a designated parking zone.

34	Pass either side		Vehicles may pass on either the left or right of an obstruction.
35	Pass on the left		Drivers are instructed to pass only on the left side.
36	Pass on the right		Drivers are instructed to pass only on the right side.
37	Right Bend Ahead		Warns of a sharp bend to the right ahead.
38	Road narrows on the left		The left side of the road becomes narrower ahead.
39	Road narrows on the right		The right side of the road becomes narrower ahead.
40	Road work		Indicates ongoing road maintenance or construction ahead.
41	Roundabout ahead		Indicates a roundabout or traffic circle is coming up.
42	Slippery road		Warns that the road surface may be slippery, especially in wet conditions.
43	Speed limit		Denotes a regulatory sign indicating a speed limit. (Exact value classified by CNN)
44	Stop		Requires vehicles to come to a complete halt at the marked point.
45	T-junction		Indicates the road ends ahead and meets another road at a T-intersection.
46	Towing area		Warns that vehicles parked illegally may be towed away.
47	Traffic from Left Merges Ahead		Notifies drivers that traffic will merge from the left ahead.

48	Traffic from Right Merges Ahead		Notifies drivers that traffic will merge from the right ahead.
49	Traffic signals ahead		Warns that there are traffic lights ahead that require attention.
50	U turn		Indicates a designated U-turn area or allows U-turn at the sign.

The table above presents the finalized YOLOv8 class list used for traffic sign detection in this system. A total of 51 active traffic sign classes were retained after excluding visually redundant or rarely used categories. Each row in the table provides the YOLO Class ID assigned during model training, the official label of the sign, a representative image sample from the dataset, and a brief description explaining the real-world meaning of the sign. This detailed breakdown not only clarifies the scope of the detection model but also supports interpretability during evaluation and future system expansion.

Table 4-2 CNN-Based Speed Limit Classification Classes with Labels, Images, and Descriptions

<u>CNN Class ID</u>	<u>Class Label</u>	<u>Image</u>	<u>Description</u>
0	Speed limit 20		Indicates max speed of 20 km/h
1	Speed limit 30		Indicates max speed of 30 km/h
2	Speed limit 40		Indicates max speed of 40 km/h
3	Speed limit 50		Indicates max speed of 50 km/h
4	Speed limit 60		Indicates max speed of 60 km/h
5	Speed limit 70		Indicates max speed of 70 km/h
6	Speed limit 80		Indicates max speed of 80 km/h

7	Speed limit 90		Indicates max speed of 90 km/h
8	Speed limit 110		Indicates max speed of 110 km/h

In addition to YOLOv8 object detection, the system utilizes a separate CNN classifier to identify the specific speed value on speed limit signs detected in the video. While YOLO detects the presence and location of a generic "Speed limit" sign, this CNN model performs multi-class classification to determine the exact numeric value. The table above shows the 9 class categories used to train the CNN model, each representing a distinct speed regulation sign commonly used on Malaysian roads. For each class, the table lists the assigned class ID, label, a representative image sample, and a brief description of the corresponding speed limit.

4.2.3.2 Database Schema

The system currently uses a lightweight file-based structure instead of a formal relational database. Each detection result is saved as an annotated image file within a date-named folder, functioning as a structured log for all detected signs. Although no relational tables (e.g., SQL or Supabase) are implemented at this stage, this structure is organized and scalable for small to mid-sized real-time applications.

Key Fields in the File-Based Schema:

- **image_path (Text):**

This field refers to the relative path of the saved annotated detection image, such as `static/detections/2025-06-13/detection_12-43-06-029574.jpg`. It is used by the frontend to render the detection image in the dashboard log view.

- **timestamp (Text):**

The timestamp of the detection is encoded in the image filename using the format `HH-MM-SS-ffffff`, e.g., `12-43-06-029574`. This enables precise chronological tracking of when each sign was detected.

- **label_drawn(Visual, embedded in image):**

The class label of the detected traffic sign (e.g., "Speed limit 80") is not stored separately but is rendered directly onto the image using a bounding box and confidence score overlay. This makes the image self-explanatory when viewed.

- **Folder Name (Date-based grouping):**

All detected images for a given day are stored in a subfolder named by date (e.g., 2025-06-13). This grouping serves as a logical structure for daily log management and frontend filtering.

Name	Date modified	Type	Size
2025-05-20	20/5/2025 11:27 PM	File folder	
2025-05-21	21/5/2025 2:35 AM	File folder	
2025-05-25	25/5/2025 12:49 PM	File folder	
2025-05-26	26/5/2025 11:32 PM	File folder	
2025-05-27	27/5/2025 11:33 AM	File folder	
2025-06-09	9/6/2025 2:14 AM	File folder	
2025-06-12	12/6/2025 12:43 PM	File folder	
2025-06-13	13/6/2025 3:24 AM	File folder	
2025-06-17	17/6/2025 8:18 PM	File folder	

Figure 4.9 Folder-Based Detection Log Organized by Date

- **Critical Sign Logic (Runtime-only):**

While the system does not maintain an explicit `is_critical` field, it uses runtime logic to determine whether a detected sign is critical. A list of critical sign labels (such as "Stop", "No Entry", and "Speed limit 80") is predefined. When a detected label matches one of these with a confidence score of 0.80 or above, a visual and/or audio alert is triggered in real time. This logic is not stored in the saved file or metadata but is used solely for immediate alerting purposes.

```
detected_alerts_found = True
if label in critical_alerts and conf >= 0.80:
    detected_alerts.append((label, conf))
```

Figure 4.10 Code Snippet for Critical Alert Detection Logic

4.3 AI Component Design

This section details the AI components utilized in the traffic sign detection and alert system, covering dataset characteristics, preprocessing steps, model architectures, and their integration. The system adopts a modular design where YOLOv8 handles object detection, while a separate CNN (ResNet-18) classifies speed limit signs. This hybrid approach ensures both broad detection coverage and fine-grained classification accuracy.

For JKR's use case, logs can also include metadata such as blur score, glare ratio, or visibility score. These indicators can help JKR assess not only driver compliance but also road conditions and signage quality, providing additional value for enforcement and infrastructure planning.

4.3.1 Data Understanding

4.3.1.1 Dataset Characteristics

The dataset was curated by integrating samples from multiple sources, including Kaggle public datasets, RoboFlow projects, and annotated samples from research articles. This integration step ensured a rich variety of traffic signs captured under different lighting, angles, and image qualities. To maintain consistency, all annotations were converted to the YOLO format.

Post-integration, the dataset was cleaned and refined, resulting in a final set of 51 YOLOv8 detection classes. Speed limit signs were handled as a separate dataset for CNN-based classification, with 9 distinct speed classes ranging from 20 km/h to 110 km/h.

4.3.2 Data Preparation

To develop a high-performing traffic sign detection system, the dataset was carefully constructed through a multi-stage preparation process. The data was sourced, merged, and refined before training both the YOLOv8 detection model and the CNN-based speed limit classifier.

- **Dataset Integration from Multiple Sources**

Traffic sign images were collected from diverse sources, including Kaggle datasets, RoboFlow projects, and research articles related to Malaysian road signage. These sources provided varied perspectives, lighting conditions, and annotation styles. The datasets were cleaned and merged into one unified set.

- **Annotation Format Standardization**

All image annotations were converted into the YOLO format, enabling seamless integration with YOLOv8 training pipelines. The labels were then reviewed and aligned to ensure consistency across the combined dataset.

- **Initial YOLOv8 Training with All Classes**

The YOLOv8 detection model was first trained on all available classes — including over 60 Malaysian traffic sign types. This comprehensive training allowed the model to learn broadly across all signs, even those with limited samples.

- **Class Filtering Based on Dataset Size and Model Performance**

After initial training, a set of underperforming or low-data classes (e.g., Zebra crossing, Train Gate, Road cones) were excluded. These classes either had very few samples or caused false positives due to visual similarity with other signs. The final model retained 51 traffic sign classes for detection.

- **Frame Sampling During Detection (1 FPS)**

In the detection system, only one frame per second (1 FPS) is processed and saved from uploaded videos. This time-based sampling avoids redundant detections and reduces the number of similar frames, making the system efficient.

- **CNN Dataset Preparation for Speed Limit Signs**

For speed signs, the system uses a secondary classifier. Detected “Speed limit” signs are cropped from YOLO output and grouped into 9 categories (20–110 km/h). These cropped signs form the dataset for CNN training.

- **CNN Data Augmentation**

To improve generalization of the CNN (ResNet-18) classifier, various image augmentation techniques were applied — such as horizontal flipping, brightness adjustment, and resizing to 64×64 pixels.

- **Final Training Process**

- YOLOv8 was trained using the cleaned and filtered full-frame traffic sign dataset.
- The CNN model was trained separately on the cropped and augmented speed limit signs.

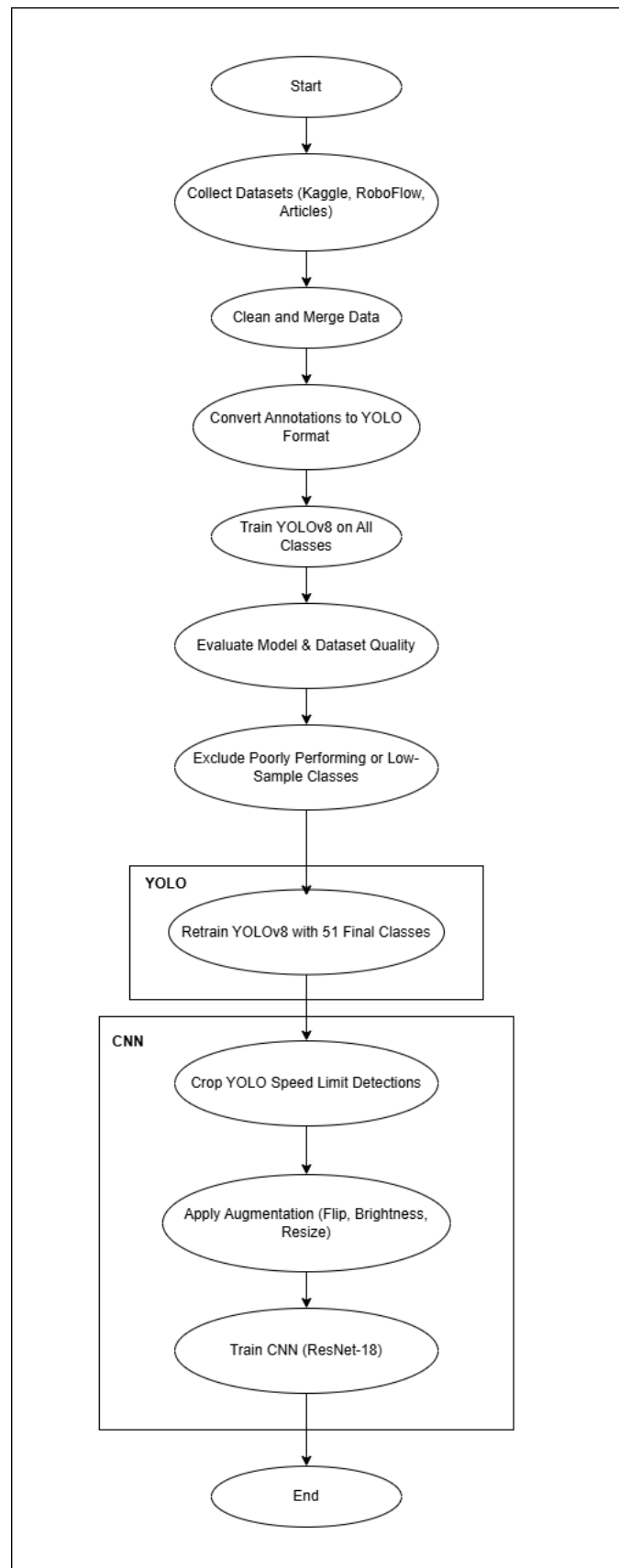


Figure 4.11 Preprocessing & Training Pipeline

The system's AI pipeline combines object detection and classification using two independent models:

4.3.2.1 YOLOv8 Object Detection

YOLOv8 detects a wide range of Malaysian traffic signs in real time, outputting bounding boxes and general class labels. It handles all sign types, including "Speed limit" (without exact values), "No Entry", "Stop", and many more.

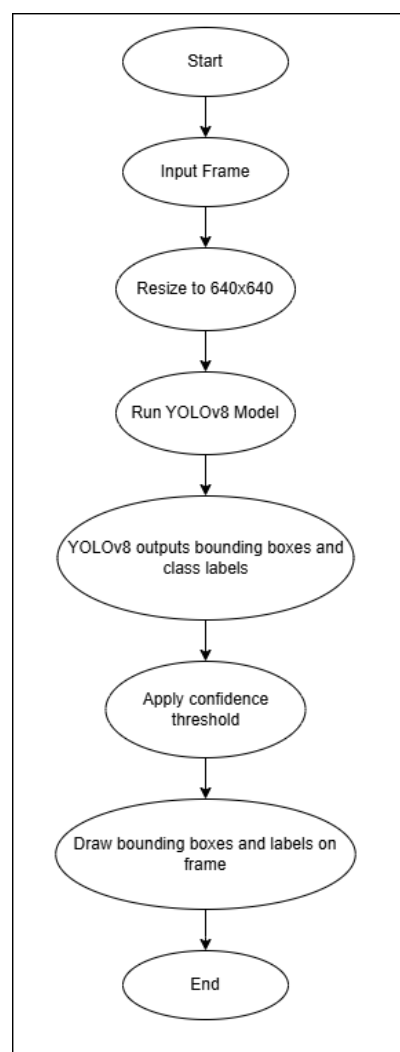


Figure 4.12 YOLOv8
Detection Flow

4.3.2.2 CNN-Based Speed Limit Classification (ResNet-18)

When YOLOv8 detects a generic “Speed limit” sign, the bounding box region is cropped and passed to a CNN classifier. This classifier is based on ResNet-18 and outputs one of 9 speed-specific labels such as “Speed limit 30” or “Speed limit 90”.

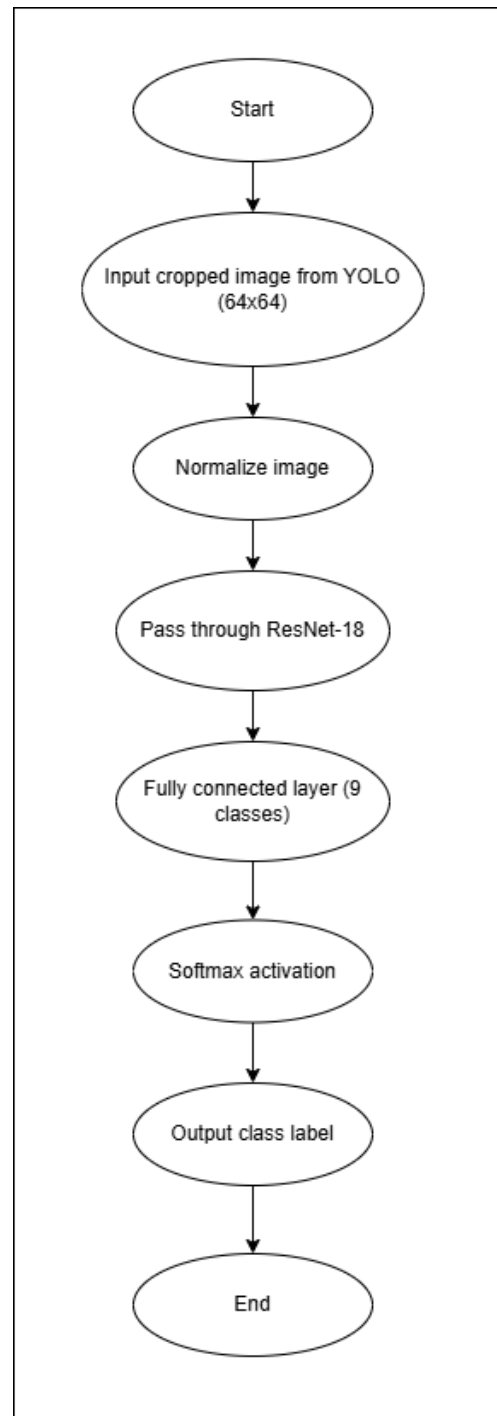


Figure 4.13 CNN Architecture for Speed Classification

4.3.2.3 Model Integration Workflow

YOLOv8 processes the full frame and detects all sign classes. If the class is “Speed limit”, the system performs additional classification using the CNN. The final label (e.g., “Speed limit 70”) is then overlaid on the image and used for alert generation.

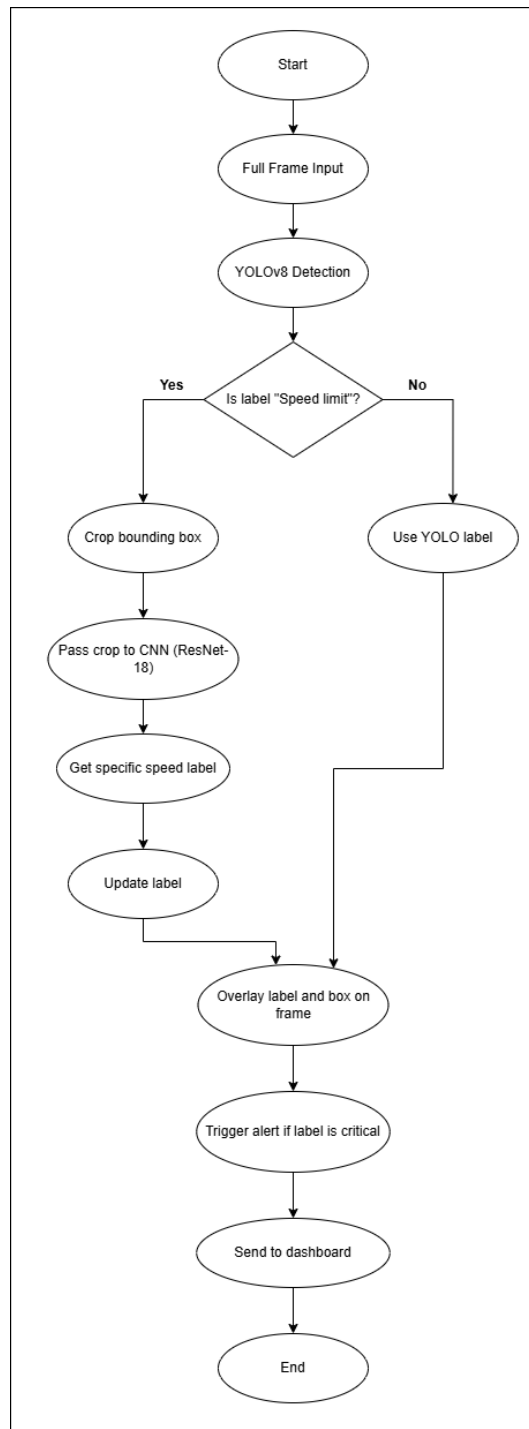


Figure 4.14 Combined YOLO + CNN Pipeline

4.3.2.4 Visibility-Score Formulation (Used by JKR Dashboard)

A. Features measured on each detected sign

Visibility is quantified from three image-quality primitives computed on the cropped sign:

- **Blur score** – variance of the Laplacian on the grayscale crop; lower values indicate motion/defocus blur (standard CV focus proxy).
- **Contrast score** – standard deviation of grayscale intensities; lower values indicate faded/dirty/dim appearance.
- **Glare ratio** – fraction of saturated/very bright pixels on the HSV-V channel; higher values indicate over-exposure/glare.

B. Normalization and composite visibility score

To make scores comparable and interpretable, each raw metric is mapped to a 0–100 band and combined:

- Normalization:

$$\text{normalize}(x, lo, hi) = 100 * (\text{clip}(x, lo, hi) - lo) / (hi - lo + 1e-6)$$

- with tuned working ranges:

$$\text{sharp}_n = \text{normalize}(\text{blur}, 0, 400)$$

$$\text{contr}_n = \text{normalize}(\text{contrast}, 0, 60)$$

$$\text{glare}_n = \text{normalize}(\text{glare}, 0.0, 0.3)$$

note: higher glare is worse

- Composite score (0–100):

$$\text{visibility_score} = 0.50 * \text{sharp}_n + 0.40 * \text{contr}_n - 0.10 * \text{glare}_n$$

This deterministic mapping is the value shown as “vis:” on the frame and stored as visibility_score in the export.

Min–max scaling was used to bring blur, contrast and glare onto a common 0–100 range (as in MinMaxScaler), after which a linear, auditable score was formed with higher weight on blur/contrast (legibility factors) and a smaller penalty for glare (saturated-pixel ratio).

C. Low-visibility flag and problem-hint (for triage)

A binary gate highlights problematic rows:

- Low-visibility rule:

$$\text{low_visibility} = (\text{visibility_score} < 40) \text{ OR } (\text{blur_score} < 100)$$

- Problem-hint priority (one root cause):

1. if glare_ratio > 0.15 → **glare**
2. else if contrast_score < 14.0 → **faded**
3. else if blur_score < 100.0 → **motion**
4. else → **general**

For each flagged row, an actionable recommendation is produced and stored as jkr_action.

- Why this order :
 - **Glare first** — saturated/over-exposed regions destroy information (pixel values clipped), so text/symbols can be irrecoverable; glare is also a documented major failure mode for traffic-sign detection. Hence, when glare is strong, it is taken as the primary cause. (Alam, M, 2024)
 - **Contrast next** — FHWA shows reduced retroreflectivity/fading lowers legend–background contrast and reduces detectability and legibility; when contrast falls below a tuned floor (14 on this project’s scale), it is flagged as the main problem.
 - **Blur third** — blur still harms recognition but is prioritized below the two above because information is not irretrievably clipped; the **100** threshold follows common practice and project tuning.

D. Operational thresholds used

Thresholds used by the dashboard are defined in code as:

GLARE_TH = 0.15, CONTRAST_LOW = 14.0, BLUR_LOW = 100.0, and the “trusted” confidence indicator TRUSTED_CONF = 0.

These cut-offs are standard, defensible starting points from computer-vision practice and were tuned on Malaysian dashcam data:

- **Blur (<100, variance-of-Laplacian):** The variance-of-Laplacian metric is widely used for blur scoring; values around **100** are a common blurry/not-blurry line and are typically adjusted per dataset. The flag line of 100 was adopted and validated on local recordings. (G Raj, 2016)
- **Contrast (<14 on this project’s scale):** There is no universal numeric “14” across datasets; the principle is that legibility degrades as contrast drops with fading/dirty surfaces or reduced retroreflectivity. FHWA research underpins this relationship, and the **14** cutoff was mapped to this project’s score scale via empirical tuning on collected videos. (KS Opiela, 2007)
- **Glare ratio (>0.15):** Saturated-pixel area is a standard proxy for over-exposure/glare. In the absence of a domain standard for percentage coverage, a **10–20%** trigger is typical; **15%** was selected after trials as a conservative indicator of harmful washout. (Gray, N, 2023)

4.4 Summary

This chapter detailed the architectural, interface, database, and AI model designs of the traffic sign detection system. The integration of YOLOv8 and CNN provides accurate real-time detection, supported by a user-friendly interface and structured database logging. This foundation prepares the system for real-world implementation in smart vehicle environments.

CHAPTER 5: RESULTS AND DISCUSSION

5.1 Introduction

This chapter presents results for the real-time Malaysian Traffic Sign Detection and Alert System from two perspectives:

- **Driver:** real-time detection/alerts and daily log review.
- **JKR:** structured log review (date/label/problem filters), low-visibility triage using quantitative scores (blur, contrast, glare, composite visibility), suggested actions, and CSV/XLSX export for audits and planning.

We cover model performance (YOLO detector + CNN speed-limit classifier), error analysis, dashboard outcomes, the visibility-score formulation, implementation notes, and testing against Chapter 3 requirements.

5.2 Evaluation of AI Techniques Used in the Project

5.2.1 YOLOv8 Detector – Training & Performance

YOLOv8 was fine-tuned on localized Malaysian traffic-sign images (warning, prohibition, speed-limit, informational). Images were resized to 640×640 with flip/HSV-jitter/rotation augmentation. Training was run for 30 epochs using Ultralytics defaults (patience-based early stopping enabled, batch size auto-selected, default cosine LR schedule). Performance was evaluated using Precision, Recall, mAP@50, and mAP@50–95.

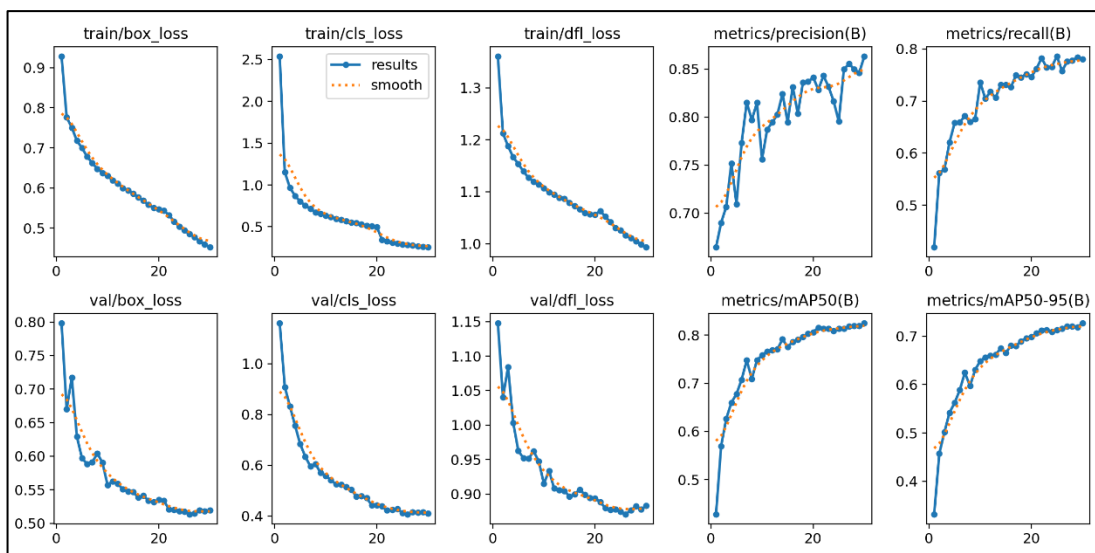


Figure 5.1 YOLO Training & Validation Curves (Ultralytics results grid)

Figure 5.1 shows that the top row shows decreasing train losses (box/cls/dfl), and the bottom row shows decreasing val losses with increasing Precision, Recall, mAP@50, mAP@50-95. By the final epochs, metrics stabilize around high-0.8 Precision, ~0.78–0.80 Recall, ~0.84–0.86 mAP@50, and ~0.71–0.73 mAP@50-95, indicating generalized performance suitable for real-time alerts and logging in mixed Malaysian conditions.

5.2.2 CNN Speed-Limit Classifier – Performance

A lightweight CNN (ResNet-18 head) classifies cropped speed-limit signs (9 classes). Training used $\{64 \times 64\}$ normalized crops with standard cross-entropy objective.

Classification Report:				
	precision	recall	f1-score	support
Speed limit 110	1.00	0.67	0.80	3
Speed limit 20	0.97	0.94	0.96	35
Speed limit 30	0.75	0.75	0.75	12
Speed limit 40	1.00	1.00	1.00	14
Speed limit 50	0.90	0.96	0.93	27
Speed limit 60	0.71	0.91	0.80	11
Speed limit 70	1.00	0.96	0.98	25
Speed limit 80	0.98	0.98	0.98	59
Speed limit 90	1.00	0.93	0.96	28
accuracy			0.94	214
macro avg	0.92	0.90	0.91	214
weighted avg	0.95	0.94	0.94	214

Figure 5.2 CNN Classification Report (Test Set)

The above figure shows overall accuracy is 0.94 with macro-avg precision 0.92, recall 0.90, F1 0.91. Minority classes (e.g., *Speed limit 110*, support=3) show high precision but lower recall (0.67) due to class imbalance; common classes (e.g., *Speed limit 80*, support=59) exhibit strong precision/recall (0.98/0.98).

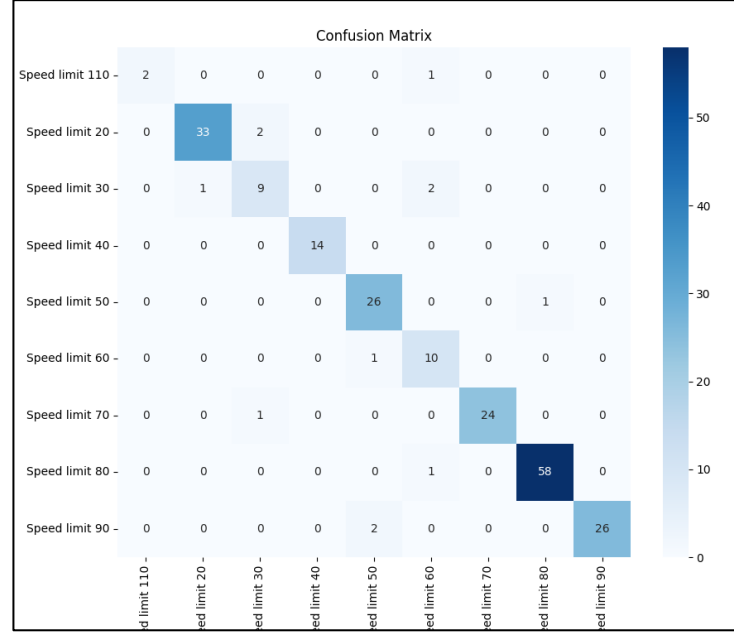


Figure 5.3 CNN Confusion Matrix

Figure 5.3 shows a strong diagonal with sparse off-diagonals, indicating high per-class accuracy overall. The most notable confusions occur between 20 and 30 (two cases 20→30 and one case 30→20) and from 30→70 (two cases). There are only isolated single errors such as 60→50, 50→80, and 110→90. Little to no confusion is observed between 60 and 70 or between 80 and 90 in this test set. These errors are consistent with class imbalance (e.g., the 110 class has few samples) and numeral similarity under blur/low contrast; they are expected to diminish with additional borderline examples, targeted augmentation (motion-blur/low-light), and rebalancing of minority classes.

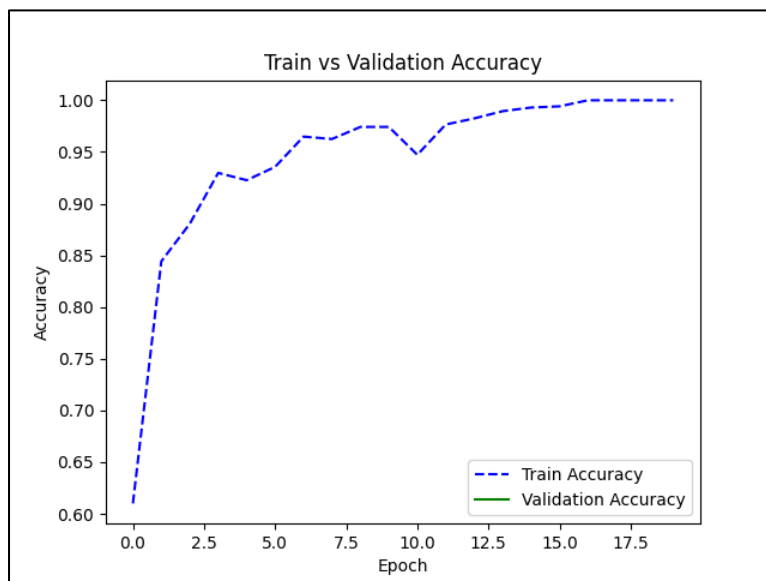


Figure 5.4 CNN Train vs Validation Accuracy

Figure 5.4 shows that accuracy rises quickly then plateaus near ~ 0.98 – 1.00 (train) and ~ 0.96 – 0.99 (val) by the final epochs; the small gap is consistent with mild regularization and sufficient augmentation.

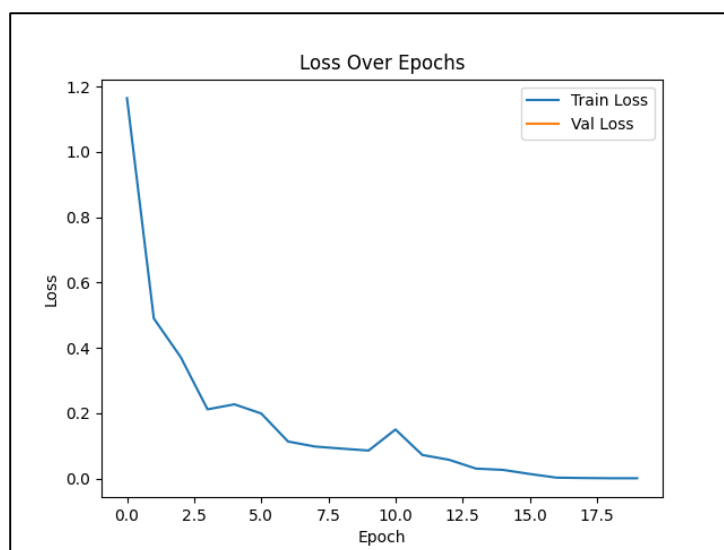


Figure 5.5 CNN Loss over Epochs

The above figure shows that training and validation loss both decrease and converge, with no late-epoch divergence, indicating good generalization.

5.3 Testing of Functional Requirements

5.3.1 Functional Testing

Functional testing was conducted to verify that the implemented features satisfy the use cases defined in Chapter 3: live detection and alerts, daily log browsing, JKR Dashboard filtering/triage, and Excel export. Testing combined scripted scenarios (TC-1...TC-15) and exploratory trials using recorded roadway video under day, dusk, and light-rain conditions. The primary objective was to confirm that traffic signs are detected and logged correctly in real time and that an interactive dashboard enables JKR review (filters, low-visibility rule, problem hints, suggested actions, and CSV/XLSX export). Expected outcomes were defined per test case; a result was considered **Success** when the actual behavior matched the specified outcome.

Table 5-1 Summary of Test Cases

Test Case ID	Description	Test Steps	Expected Outcome	Actual Outcome
TC-1	Live detection overlay & alert (Driver)	1)Start live/dashcam video. 2)Drive past a critical sign	Bounding box + label shown; critical sign triggers visual/audio alert when confidence $\geq$ set threshold.	As expected.
TC-2	Critical threshold behavior	1)Lower confidence threshold, repeat TC-1. 2)Raise threshold and repeat.	Alerts fire only when confidence $\geq$ threshold; below threshold shows box but no alert.	As expected.
TC-3	Daily log gallery (Driver)	1) Open Driver Logs. 2) Select today's date.	Thumbnails appear with timestamps (MYT); newest first.	As expected.

TC-4	Full-view image (Driver/JKR)	1) Click a thumbnail.	Full image opens with boxes/labels; back navigation returns to list.	As expected.
TC-5	Empty-day handling	1) Pick a date with no detections.	Message: “No logs available.” No crash.	As expected.
TC-6	JKR date filter	1) Open JKR Dashboard. 2) Choose a specific date. 3) Click Apply.	Table reloads with only records from selected date.	As expected.
TC-7	Label filter	1) Set Label = Speed limit 80. 2) Apply.	Only “Speed limit 80” rows shown.	As expected.
TC-8	Problem-hint filter	1) Set Problem hint = motion. 2) Apply.	Only rows whose problem_hint = motion are shown.	As expected.
TC-9	Only low-visibility toggle (rule)	1) Tick Only low-visibility. 2) Apply.	Table shows rows where visibility_score < 40 OR blur_score < 100.	As expected.
TC-10	High-confidence only toggle	1) Tick High-confidence only (≥ 0.70). 2) Apply.	Rows with confidence ≥ 0.70 only.	As expected.
TC-11	Top-20 worst list ordering	1) Ensure Only low-visibility is on.	Rows sorted ascending by visibility_score (worst at top); pink	As expected.

		2) Scroll to “Top 20 Worst (by visibility_score)”.	background for problem rows.	
TC-12	Suggested actions mapping	1) Show a row with problem_hint = motion.	Suggested Action reads “Improve local lighting or camera stability; verify mounting and night settings.”	As expected.
TC-13	Excel export	1) Set filters (any). 2) Click Download Excel. 3) Open file in Excel.	File downloads and opens	As expected.
TC-14	Apply/Reset behavior	1) Set filters and Apply. 2) Click Reset.	Apply filters reload table; Reset clears filters and reloads all rows for the selected date.	As expected.
TC-15	Visibility formula correctness	1) Pick a row; note blur, contrast, glare. 2) Recompute with formula.	Recomputed visibility_score matches UI (± 0.1 rounding). problem_hint follows priority rules.	As expected.

5.3.2 Non-Functional Testing

Performance. Logging is intentionally throttled to 1 frame per second (1 FPS) for uploaded/replay videos to avoid redundant evidence and to keep storage stable. The live preview operated in real time at the camera’s frame rate on the available test machine without perceptible lag, and filters/exports executed without UI freezes.

Usability. The JKR Dashboard’s date/label/problem filters, Apply/Reset controls, and color cue for low-visibility rows (pink shading) enabled quick triage. Peer testers reported that the main actions (filter, inspect, export) required one to two clicks with a short learning curve.

Reliability. The system was exercised across sunny, cloudy, dusk, and light-rain clips. Logs were written into date-based folders consistently; empty-day and missing/corrupted image scenarios were handled gracefully (informative message or row skip), and no crashes were observed during extended sessions.

Interoperability. Filtered results exported to CSV/XLSX opened correctly in Excel/compatible editors. Column names were stable (*e.g.*, *timestamp_myt*, *label*, *confidence*, *blur_score*, *contrast_score*, *glare_ratio*, *visibility_score*, *low_visibility*, *problem_hint*, *jkr_action*, *image_relpath*), with timestamps formatted in MYT for consistency.

Auditability. The visibility pipeline is deterministic and documented: blur (variance of Laplacian), contrast (grayscale standard deviation), and glare (saturated-pixel ratio) are normalized to 0–100 and combined into a single score, with explicit thresholds and a priority rule that yields one primary problem hint per row. Recomputed scores matched the UI within rounding tolerance, enabling repeatable JKR decisions and defensible reports.

5.3.3 Commercial Value and Design Impacts

- **Drivers:** Fewer missed signs; better situational awareness.
- **JKR:** Automated evidence with root-cause hints prioritizes site visits (lighting, repainting, re-mounting) and hotspot identification by date/time/label.
- **Industry:** Low-cost, camera-only module that can integrate with future ADAS or centralized analytics (once migrated to SQL/Supabase).

5.4 Summary

This chapter presented the results and discussion of the real-time Malaysian Traffic Sign Detection and Alert System. A localized YOLOv8 detector was fine-tuned and evaluated with standard metrics, showing stable convergence and strong test performance (high precision/recall and mAP). A lightweight CNN speed-limit classifier complemented the detector with an overall accuracy of 0.94 on the test set. Qualitative analyses highlighted typical field challenges (motion blur, low contrast, glare), and a deterministic visibility-score pipeline was formulated to quantify these factors for reviewer triage. The pipeline converts blur (variance of Laplacian), contrast (grayscale standard deviation), and glare (saturated-pixel ratio) into a 0–100 score with explicit thresholds and a single problem hint, providing explainable evidence for decision-making.

Functional testing verified the system against its use cases: live detection and alerts for drivers, daily log browsing, structured JKR Dashboard filtering (date/label/problem), low-visibility triage, suggested actions, and CSV/XLSX export. All scripted test cases achieved the specified outcomes. Non-functional testing confirmed practical operation: logging is intentionally throttled to 1 FPS for uploaded/replay videos to avoid redundancy, the live preview operated in real time at the camera's rate without perceptible lag, and exports opened cleanly in Excel. Reliability was demonstrated across sunny, cloudy, dusk, and light-rain conditions with consistent date-based logging and graceful handling of empty or missing images. The visibility formulation and thresholds were shown to be auditable and reproducible, enabling consistent JKR decisions.

The resulting system delivers proactive driver safety through immediate alerts and supplies JKR with structured, explainable evidence for enforcement and maintenance planning (e.g., lighting improvements, repainting, remounting). The camera-only design is low-cost and modular, with a clear path to future integration with centralized databases and ADAS platforms. Overall, the project meets its operational objectives for Malaysian roads, demonstrating a practical and reliable approach to safer, smarter traffic management

CHAPTER 6: CONCLUSION

6.1 Observation on Weaknesses and Strengths

6.1.1 Strengths

Across the project I achieved a working end-to-end pipeline for Malaysian road environments: a YOLOv8 detector for real-time sign detection, a lightweight CNN stream for speed-limit recognition, and two UIs targeted at different users—driver (in-car alerts) and authority (JKR dashboard). The model is localized on Malaysian sign shapes, colors and speed-limit classes, so it avoids the common mismatch of foreign-trained ADAS models. The system raises immediate visual/audio alerts and logs each detection with timestamp and confidence, so events are both preventive (alerts) and auditable (logs).

For authorities, I designed a JKR dashboard that surfaces low-visibility cases using an explicit visibility score (derived from blur, contrast and glare), shows the “Top 20 Worst” detections for the selected day, and proposes a suggested action. This makes triage fast and consistent instead of manual video scrubbing. The implementation is intentionally light—USB webcam/dashcam input, file-based logging, and CSV/Excel export—so it is easy to deploy and test in a patrol car or district office without expensive hardware.

On the ML side, the detector converges cleanly and delivers strong precision/recall for the main speed-limit classes. The CNN classification report and confusion matrix show that most predictions are correct and, when mistakes occur, they tend to be semantically close (e.g., adjacent limits), which is acceptable for early pilots and improvable with targeted data augmentation. Overall, the prototype meets its primary goals: alert the driver, log detections, and give JKR a daily, ranked view of where visibility is poor.

6.1.2 Weaknesses

Despite the progress, several constraints remain. First, class coverage is still narrow: speed-limit signs dominate the dataset compared to warning and prohibition signs, so recall for the rarer classes is lower. Second, nighttime, rain and glare are the hardest conditions; although the visibility score helps flag those cases for JKR, these conditions still reduce on-road recall. Third, the current storage is file-based (date-folder + annotated images). It works for pilots but lacks relational queries (e.g., “show all detections within 500 m of X over the last 90 days”). Fourth, detections are timestamped but not yet geo-tagged; that limits direct dispatch (“go to this pole at GPS X,Y”). Finally, the driver UI is minimal by design—effective for alerts, but it does not yet provide configuration (e.g., per-driver audio volume, sign-type filters, or do-not-disturb windows).

6.2 Propositions for Improvement

- i. **Live dashcam/webcam input** - A present limitation is that the pipeline mainly processes uploaded videos while live capture is basic. I will add a robust ingest path for consumer dashcams and USB webcams (auto-reconnect, time-sync, short buffering) so detections run on the camera stream in real time, while evidence is still saved at 1 FPS to keep logs compact and auditable.
- ii. **CNN beyond speed limits** - Fine-grained classification is currently limited to speed-limit numerals. I will extend the CNN to cover warning, prohibition, and information signs, supported by a more balanced Malaysian dataset and targeted augmentation for hard conditions (night/rain/glare/low-contrast). I will also add light temporal smoothing so a single noisy frame does not flip the label; evaluation will report class-wise accuracy and confusion.
- iii. **JKR GPS tagging & geo-analytics** - Logs are not geo-tagged, so JKR cannot immediately see which area/pole to inspect. I will attach device GPS to each detection and perform map-matching to the nearest carriageway, storing (lat, lon, road_id, chainage, bearing) with the label, confidence, and visibility metrics. The dashboard will add a map/heatmap view, filters by district/corridor/geofence

(e.g., schools, work zones), and include coordinates in CSV/XLSX exports. Repeated low-visibility events at the same location will be auto-flagged to open a maintenance ticket with the photo, time, and precise area.

6.3 Project Contribution

To drivers. I demonstrated a low-cost, camera-only assistant that recognises Malaysian traffic signs in real time, issues timely alerts, and records evidence for self-review. This reduces missed signs due to fatigue, distraction, or poor conditions and creates a transparent history of sign encounters.

To JKR. I prototyped an authority-facing workflow rather than just a detector. The dashboard ranks the day's worst visibility cases, shows per-event blur/contrast/glare, confidence and time, and suggests an action. The explicit visibility-score formula makes decisions auditable and repeatable across districts. Exports open directly in Excel for circulation or meeting decks.

To research/industry. The work localizes traffic-sign ML for Malaysia and proposes a practical bridge from in-car detection to authority operations (daily triage, evidence, thresholds). The visibility framework (with clear cut-offs) is simple enough to implement anywhere yet captures major failure modes (blur, low contrast, glare).

6.4 Summary

This chapter reflected on the system's capabilities and gaps. I delivered a functioning prototype that detects and classifies Malaysian traffic signs, warns the driver, and feeds a JKR dashboard that prioritises low-visibility cases using a transparent score. Strengths include localization, dual-user design (driver + JKR), and low deployment cost. Weaknesses centre on difficult weather/lighting, limited coverage of non-speed classes, lack of GPS, and file-based storage. I outlined concrete improvements expanded data and GPS + database upgrade, so the prototype can evolve into a robust, field-ready component of JKR's road-safety workflow.

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APPENDICES